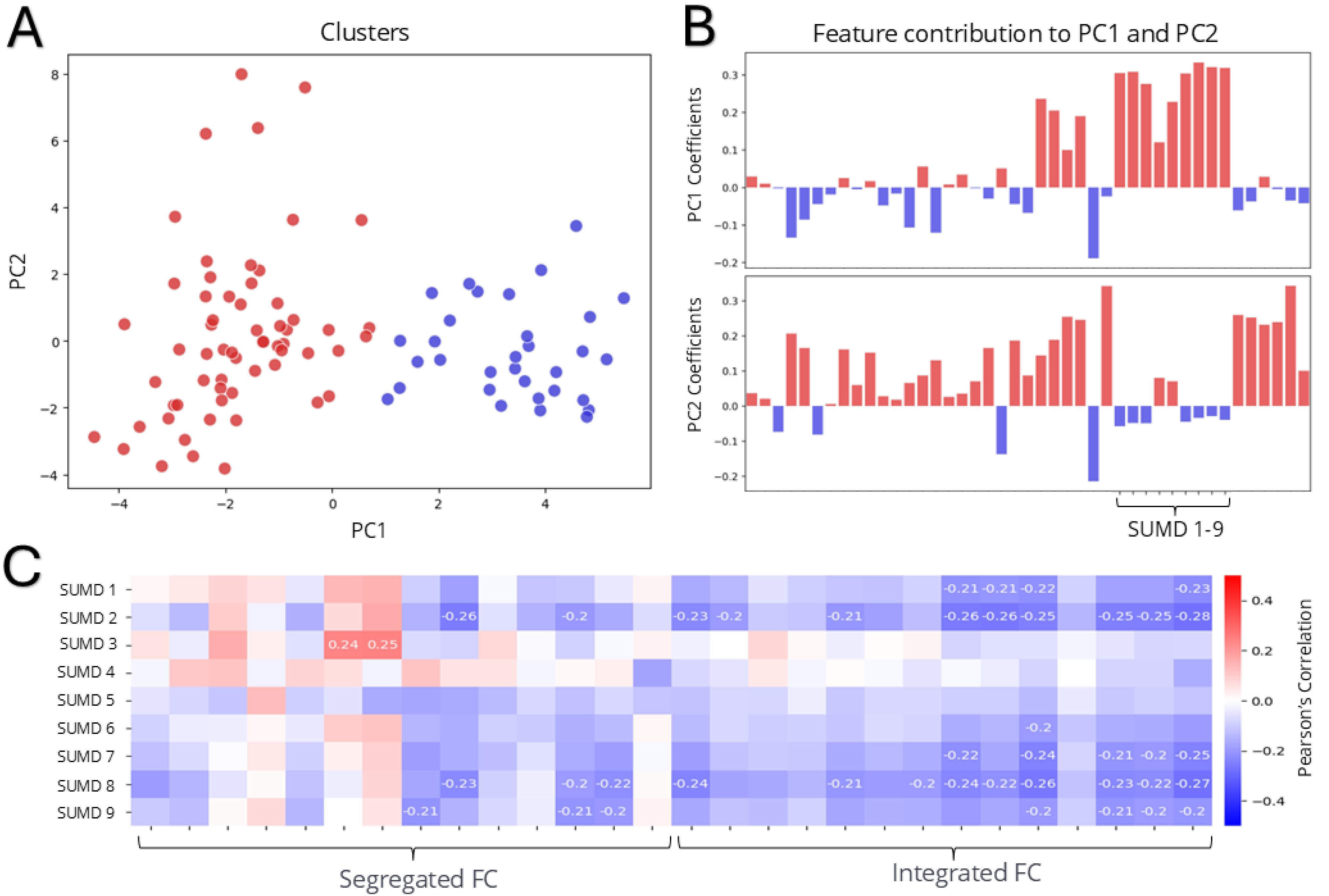
Research Updates


A
Clusters
PC1
PC2
B
Feature contribution to PC1 and PC2
PC1 Coefficients
PC2 Coefficients
SUMD 1-9
C
SUMD 1
SUMD 2
SUMD 3
SUMD 4
SUMD 5
SUMD 6
SUMD 7
SUMD 8
SUMD 9
Pearson's Correlation
Segregated FC
Integrated FC

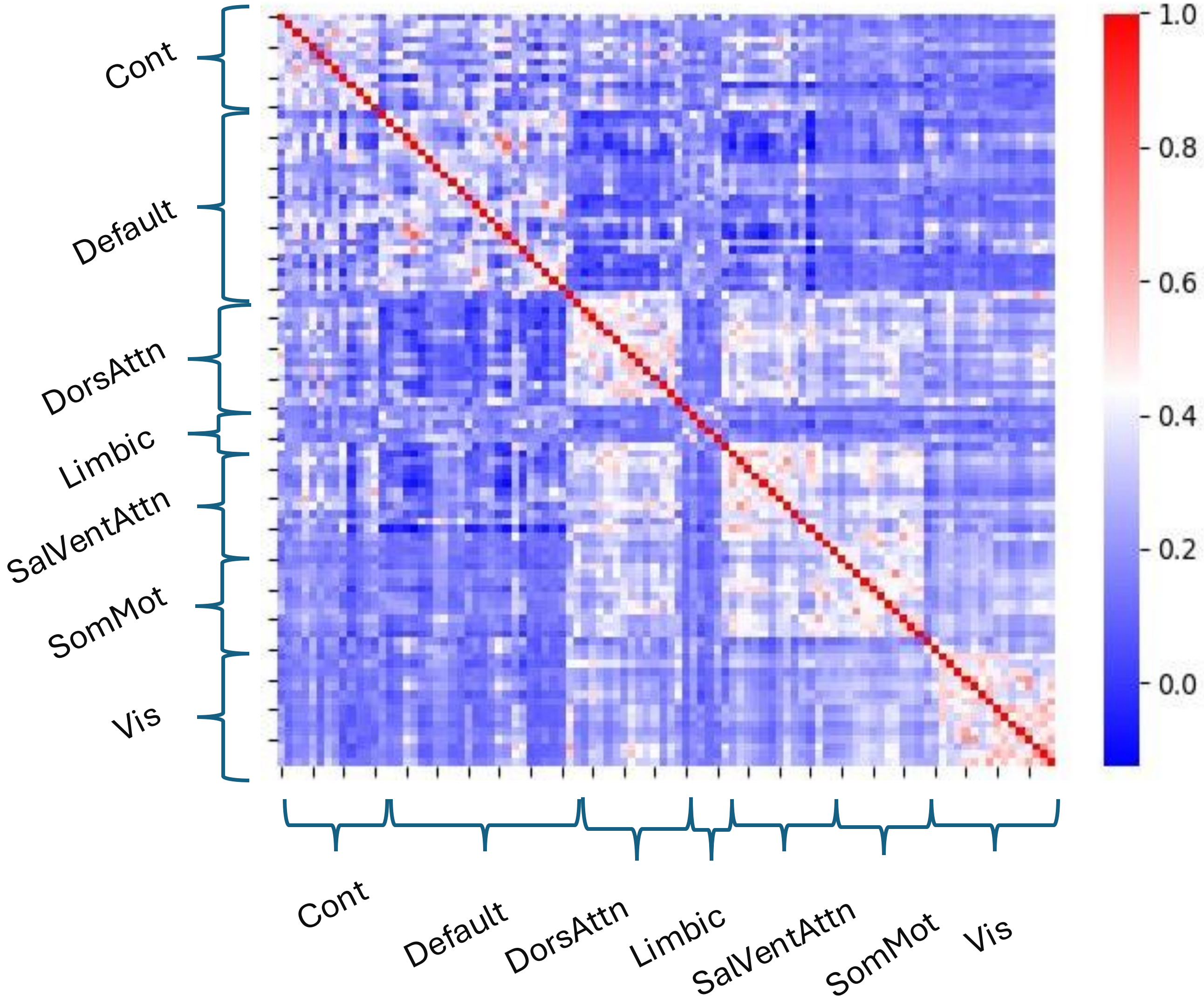
- fMRIprep processed BOLD
    - ICA – AROMA to handle motion artifacts, without mask
    - GS, WM, CSF (derivatives and quadratic terms regressed out)
- 7 Networks 100 parcellations
    - Schaefer2018_100Parcels_7Networks_regrid.nii.gz
- Checked timeseries (Error if variance < 0.01)
    - Error did not raise

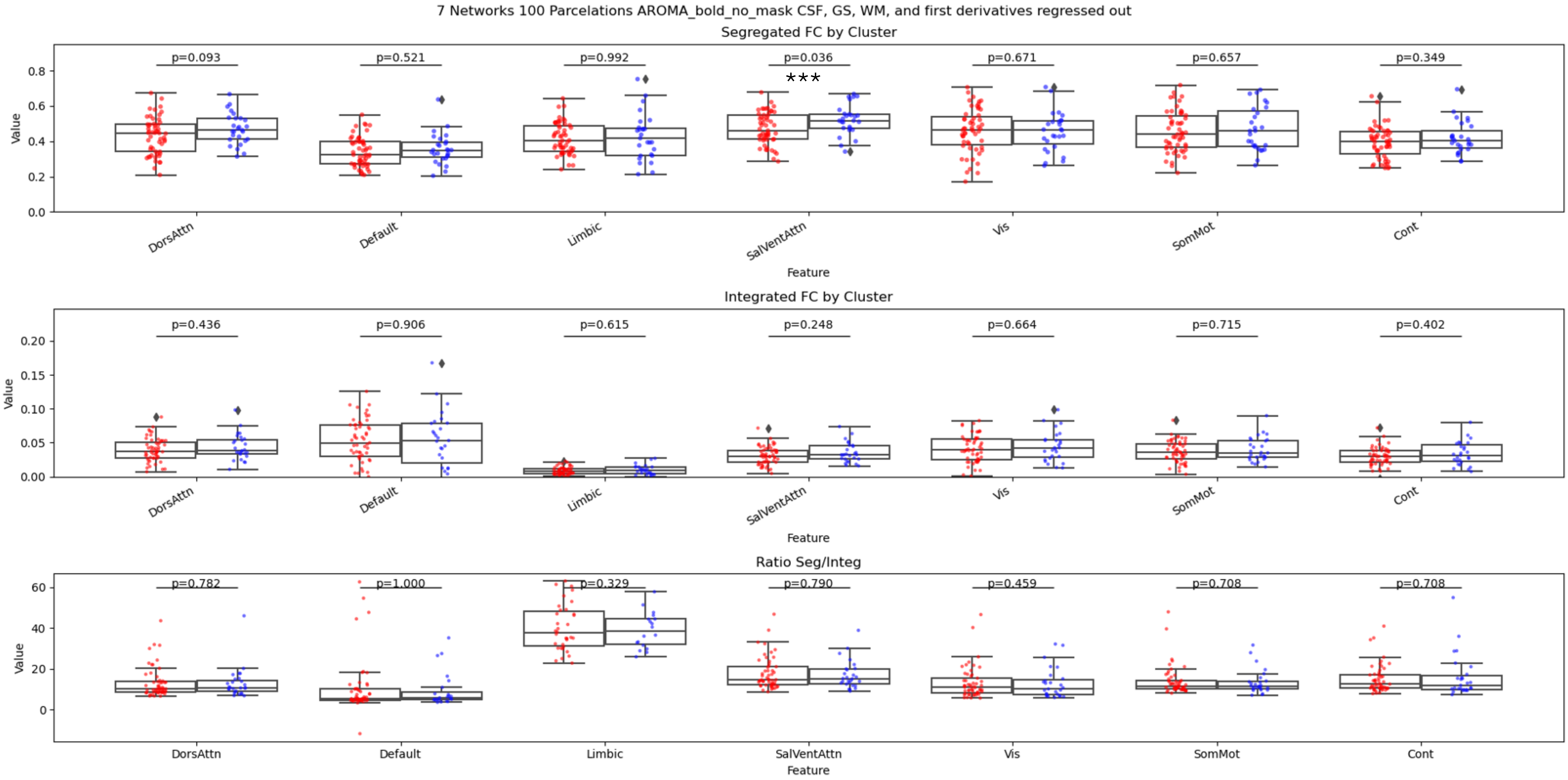
- To try: registered in MNI space

Disclaimer!

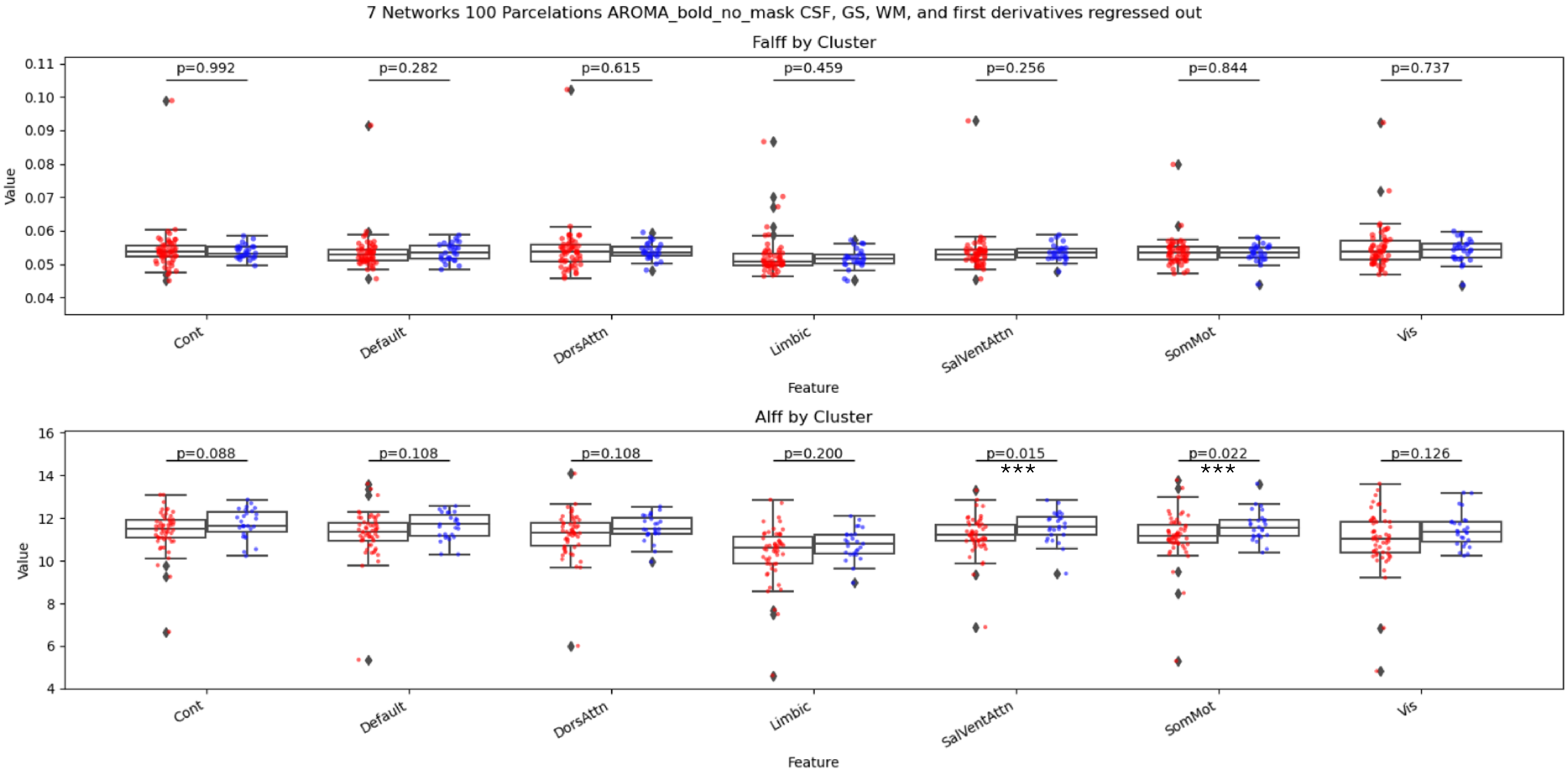
Sorry for the visualizations, it's a work in process and main focus for the following period !

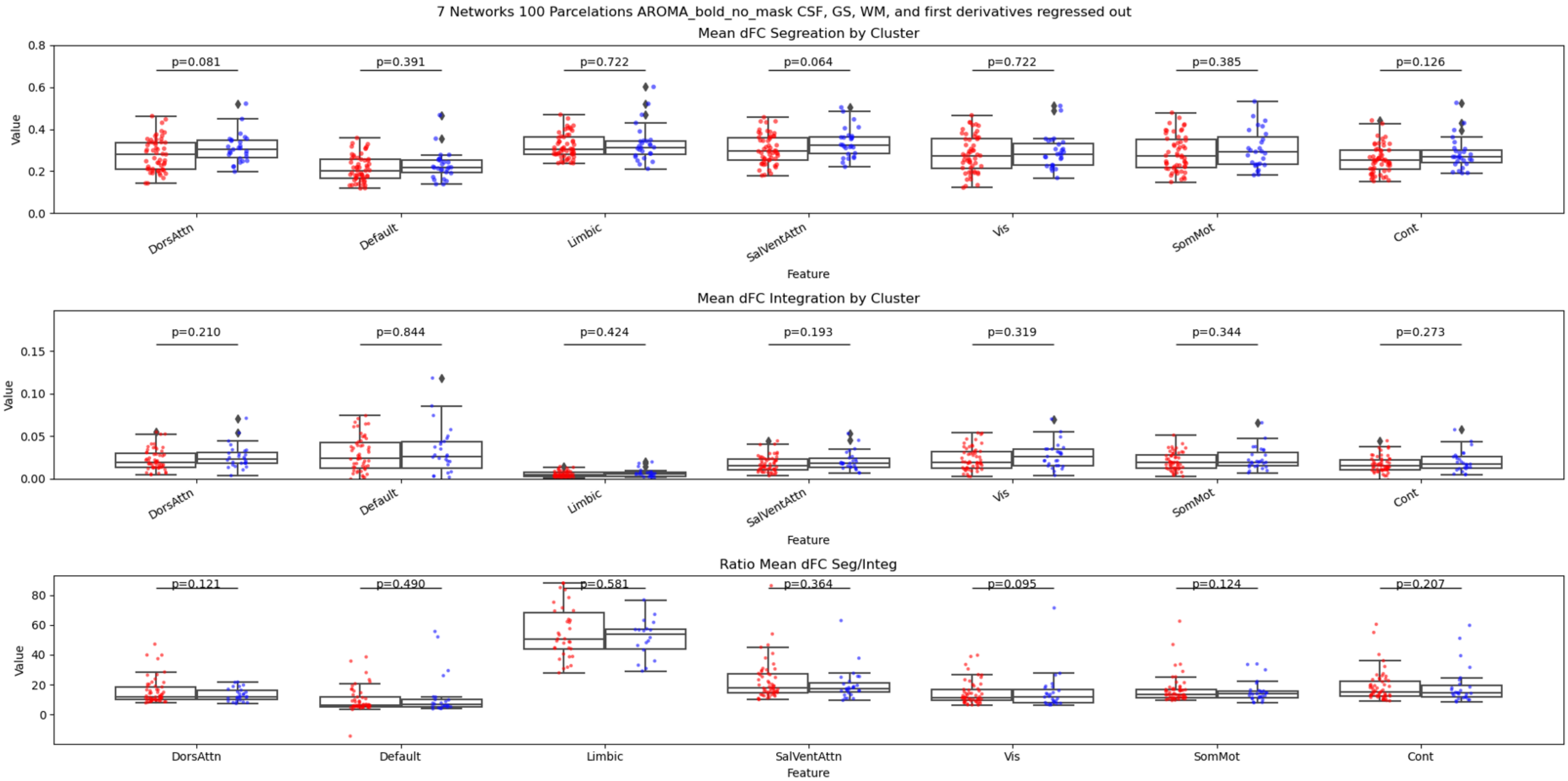
7 Networks 100 Parcellations AROMA no mask CSF, GS, WM regressed out





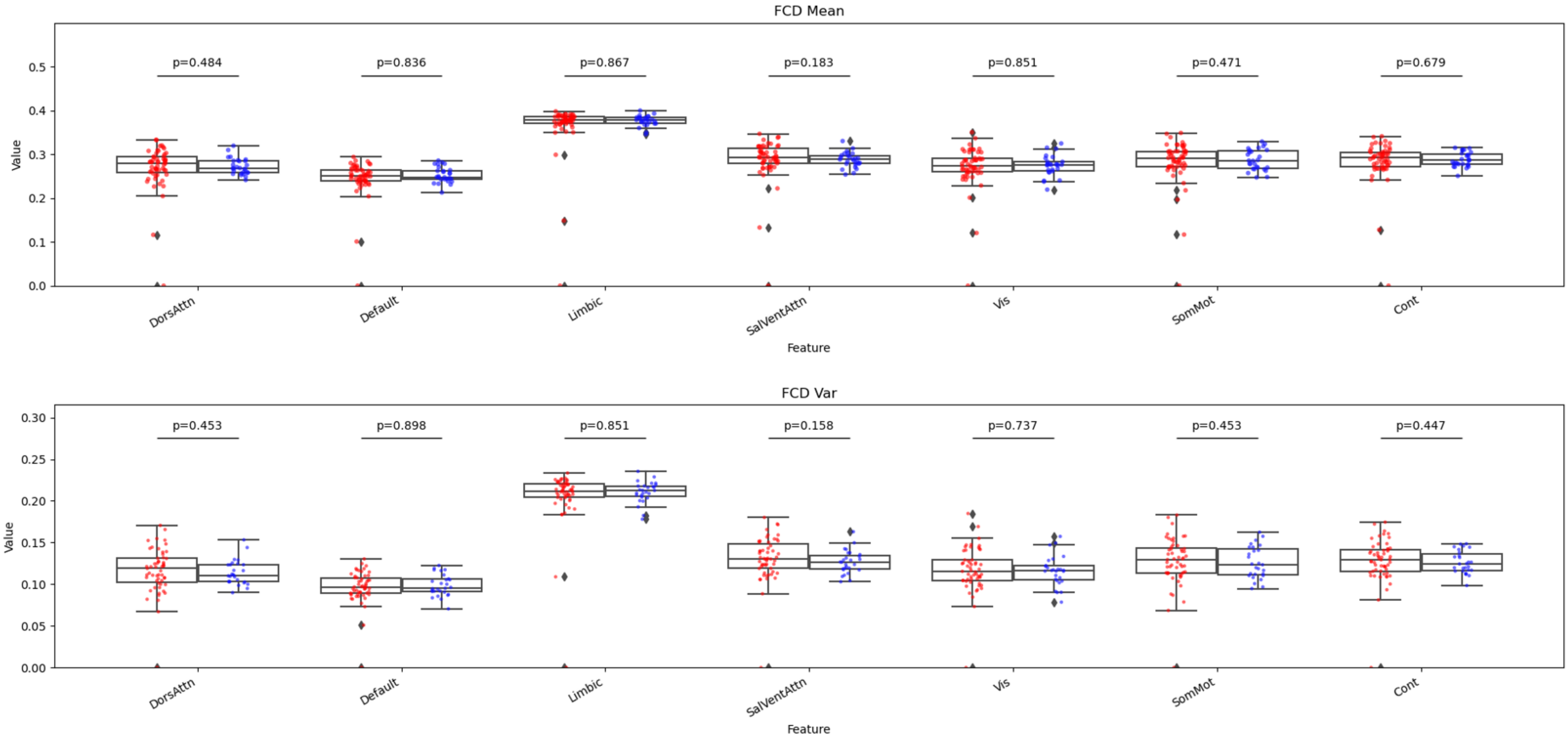
7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out
Segregated FC by Cluster
p=0.093
p=0.521
p=0.992
p=0.036
***
p=0.671
p=0.657
p=0.349
Value
0.0
0.2
0.4
0.6
0.8
DorsAttn
Default
Limbic
SalVentAttn
Vis
SomMot
Cont
Feature
Integrated FC by Cluster
p=0.436
p=0.906
p=0.615
p=0.248
p=0.664
p=0.715
p=0.402
Value
0.00
0.05
0.10
0.15
0.20
DorsAttn
Default
Limbic
SalVentAttn
Vis
SomMot
Cont
Feature
Ratio Seg/Integ
p=0.782
p=1.000
p=0.329
p=0.790
p=0.459
p=0.708
p=0.708
Value
0
20
40
60
DorsAttn
Default
Limbic
SalVentAttn
Vis
SomMot
Cont
Feature


7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out
Falff by Cluster
p=0.992
p=0.282
p=0.615
p=0.459
p=0.256
p=0.844
p=0.737
Value
Cont
Default
DorsAttn
Limbic
SalVentAttn
SomMot
Vis
Feature
Alff by Cluster
p=0.088
p=0.108
p=0.108
p=0.200
p=0.015
***
p=0.022
***
p=0.126
Value
Cont
Default
DorsAttn
Limbic
SalVentAttn
SomMot
Vis
Feature

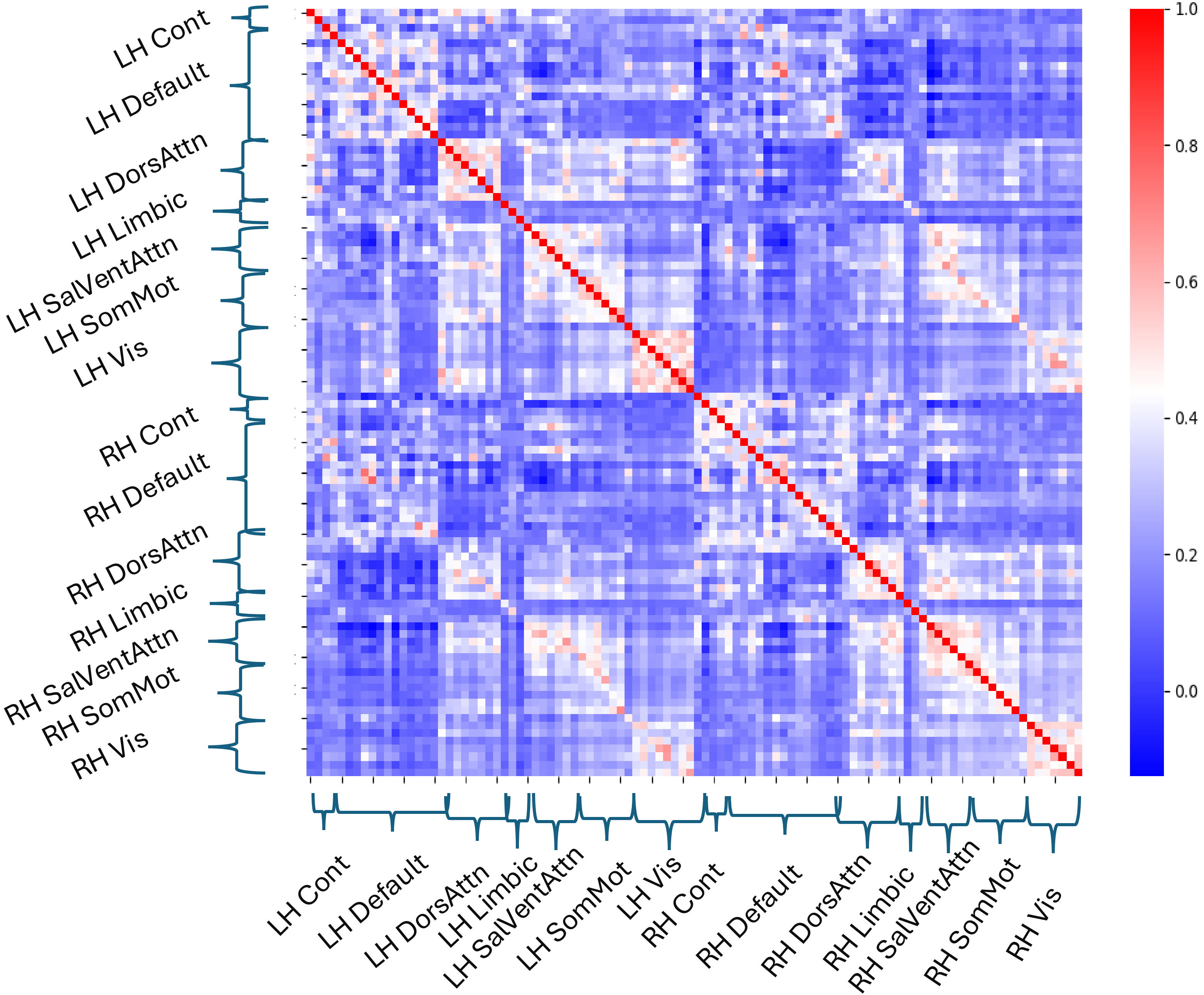


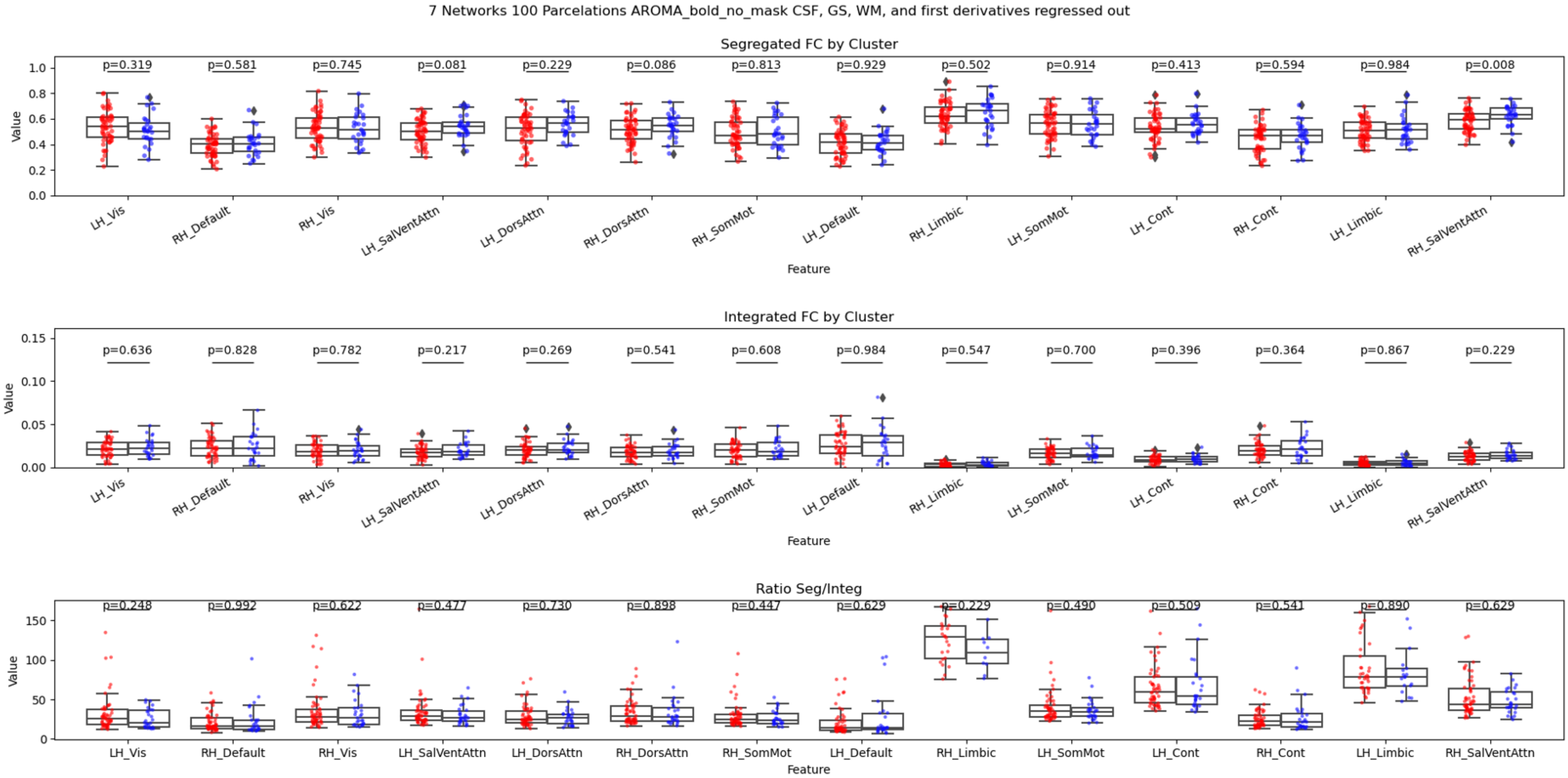
7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out
Mean dFC Segreation by Cluster
p=0.081
p=0.391
p=0.722
p=0.064
p=0.722
p=0.385
p=0.126
Value
Feature
DorsAttn
Default
Limbic
SalVentAttn
Vis
SomMot
Cont
Mean dFC Integration by Cluster
p=0.210
p=0.844
p=0.424
p=0.193
p=0.319
p=0.344
p=0.273
Ratio Mean dFC Seg/Integ
p=0.121
p=0.490
p=0.581
p=0.364
p=0.095
p=0.124
p=0.207

7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out

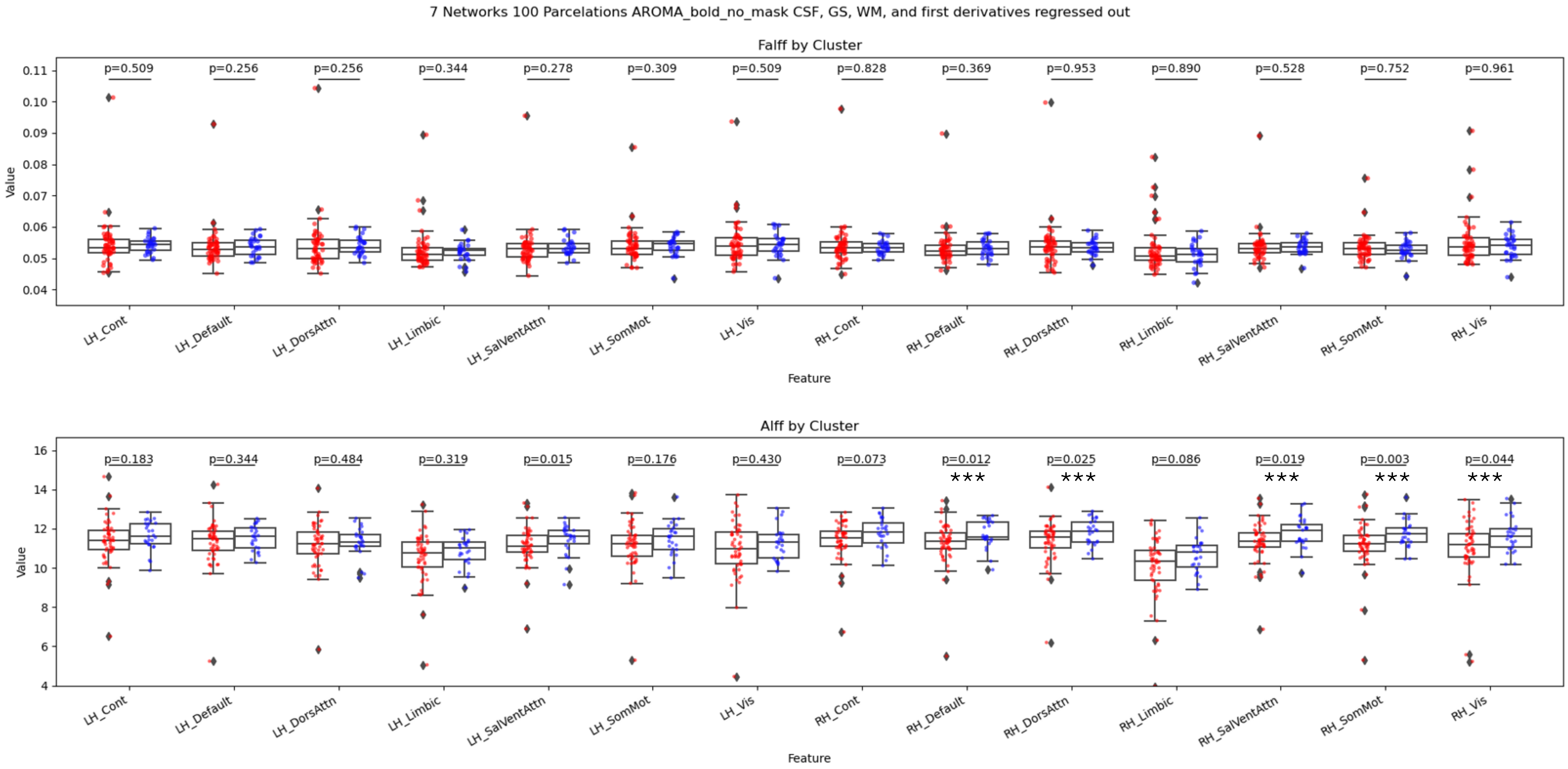


7 Networks 100 Parcellations AROMA no mask CSF, GS, WM regressed out 2 Hemispheres

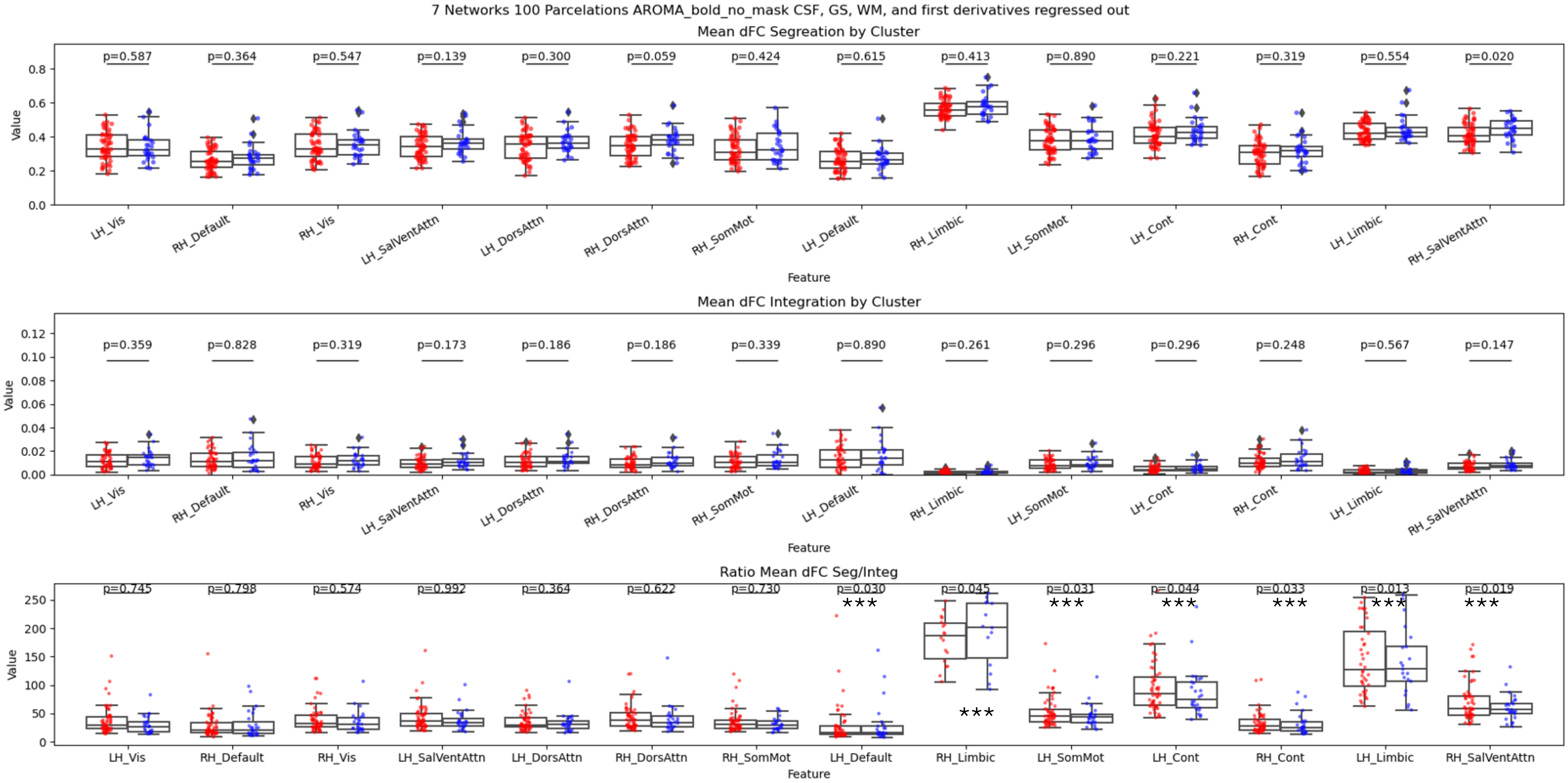




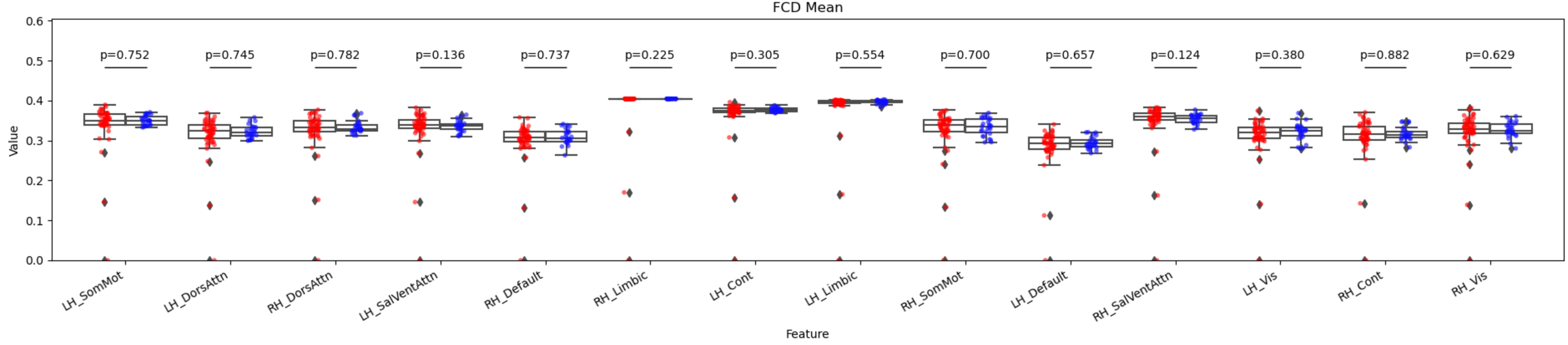
7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out
Segregated FC by Cluster
p=0.319 p=0.581 p=0.745 p=0.081 p=0.229 p=0.086 p=0.813 p=0.929 p=0.502 p=0.914 p=0.413 p=0.594 p=0.984 p=0.008
Value
Feature
LH_Vis RH_Default RH_Vis LH_SalVentAttn LH_DorsAttn RH_DorsAttn RH_SomMot LH_Default RH_Limbic LH_SomMot LH_Cont RH_Cont LH_Limbic RH_SalVentAttn
Integrated FC by Cluster
p=0.636 p=0.828 p=0.782 p=0.217 p=0.269 p=0.541 p=0.608 p=0.984 p=0.547 p=0.700 p=0.396 p=0.364 p=0.867 p=0.229
Value
Feature
Ratio Seg/Integ
p=0.248 p=0.992 p=0.622 p=0.477 p=0.730 p=0.898 p=0.447 p=0.629 p=0.229 p=0.490 p=0.509 p=0.541 p=0.890 p=0.629
Value
Feature

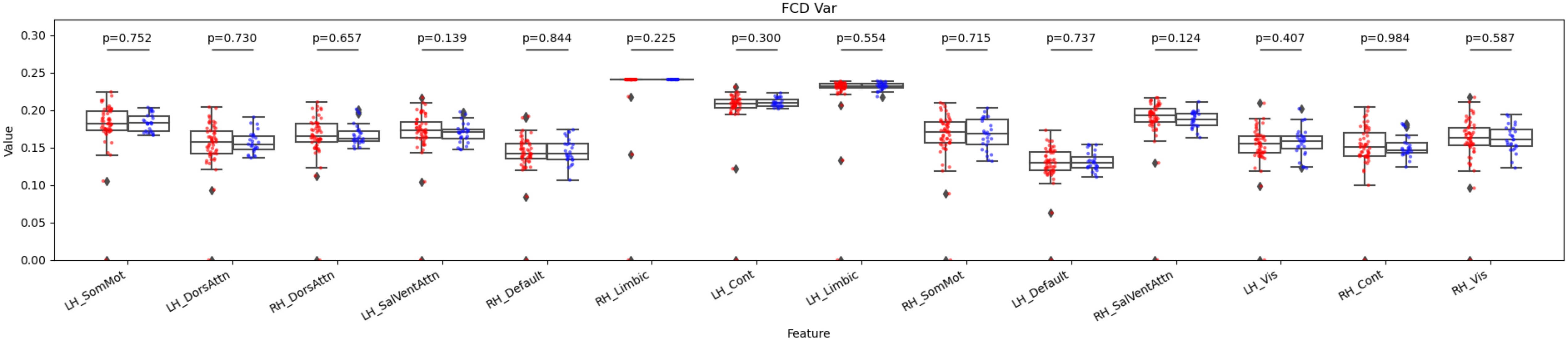


7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out
Falff by Cluster
p=0.509
p=0.256
p=0.256
p=0.344
p=0.278
p=0.309
p=0.509
p=0.828
p=0.369
p=0.953
p=0.890
p=0.528
p=0.752
p=0.961
Value
Feature
LH_Cont
LH_Default
LH_DorsAttn
LH_Limbic
LH_SalVentAttn
LH_SomMot
LH_Vis
RH_Cont
RH_Default
RH_DorsAttn
RH_Limbic
RH_SalVentAttn
RH_SomMot
RH_Vis
Alff by Cluster
p=0.183
p=0.344
p=0.484
p=0.319
p=0.015
p=0.176
p=0.430
p=0.073
p=0.012
p=0.025
p=0.086
p=0.019
p=0.003
p=0.044
***
***
***
***
***
Value
Feature

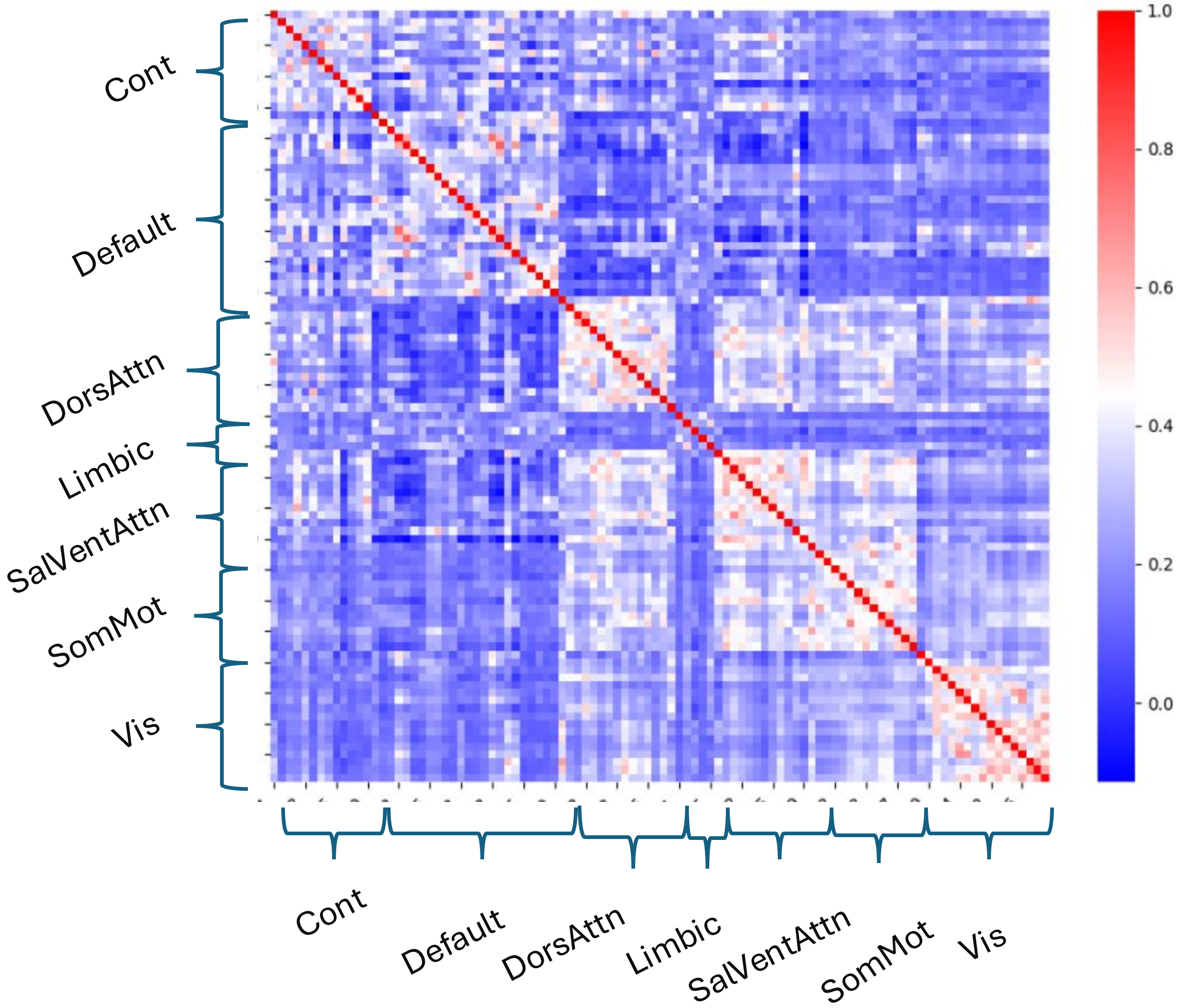

7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out
Mean dFC Segreation by Cluster
p=0.587
p=0.364
p=0.547
p=0.139
p=0.300
p=0.059
p=0.424
p=0.615
p=0.413
p=0.890
p=0.221
p=0.319
p=0.554
p=0.020
Value
0.0
0.2
0.4
0.6
0.8
LH_Vis
RH_Default
RH_Vis
LH_SalVentAttn
LH_DorsAttn
RH_DorsAttn
RH_SomMot
LH_Default
RH_Limbic
LH_SomMot
LH_Cont
RH_Cont
LH_Limbic
RH_SalVentAttn
Feature
Mean dFC Integration by Cluster
p=0.359
p=0.828
p=0.319
p=0.173
p=0.186
p=0.186
p=0.339
p=0.890
p=0.261
p=0.296
p=0.296
p=0.248
p=0.567
p=0.147
Value
0.00
0.02
0.04
0.06
0.08
0.10
0.12
LH_Vis
RH_Default
RH_Vis
LH_SalVentAttn
LH_DorsAttn
RH_DorsAttn
RH_SomMot
LH_Default
RH_Limbic
LH_SomMot
LH_Cont
RH_Cont
LH_Limbic
RH_SalVentAttn
Feature
Ratio Mean dFC Seg/Integ
p=0.745
p=0.798
p=0.574
p=0.992
p=0.364
p=0.622
p=0.730
p=0.030
p=0.045
p=0.031
p=0.044
p=0.033
p=0.013
p=0.019
***
***
***
***
***
***
***
Value
0
50
100
150
200
250
LH_Vis
RH_Default
RH_Vis
LH_SalVentAttn
LH_DorsAttn
RH_DorsAttn
RH_SomMot
LH_Default
RH_Limbic
LH_SomMot
LH_Cont
RH_Cont
LH_Limbic
RH_SalVentAttn
Feature

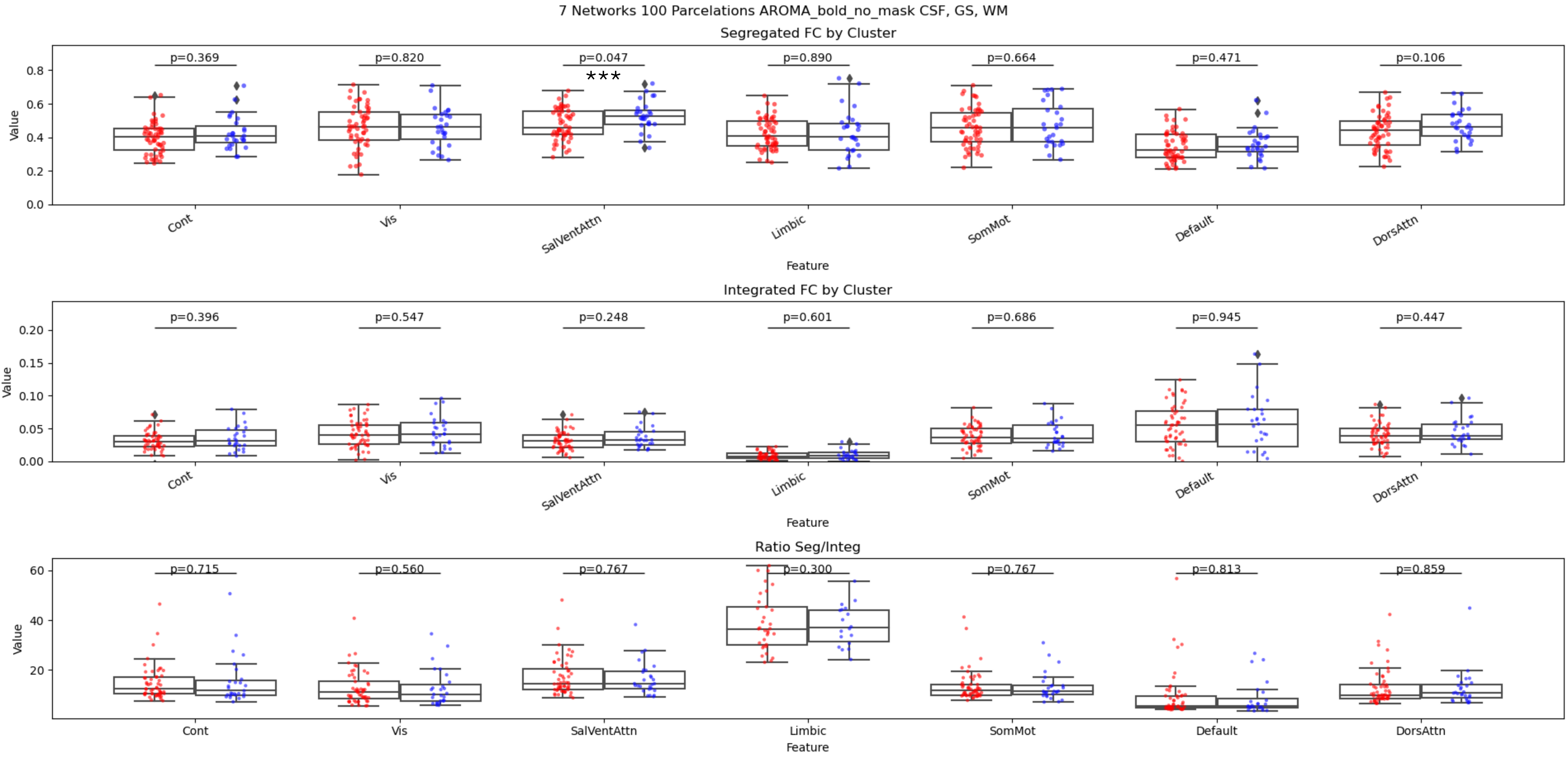
7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, and first derivatives regressed out



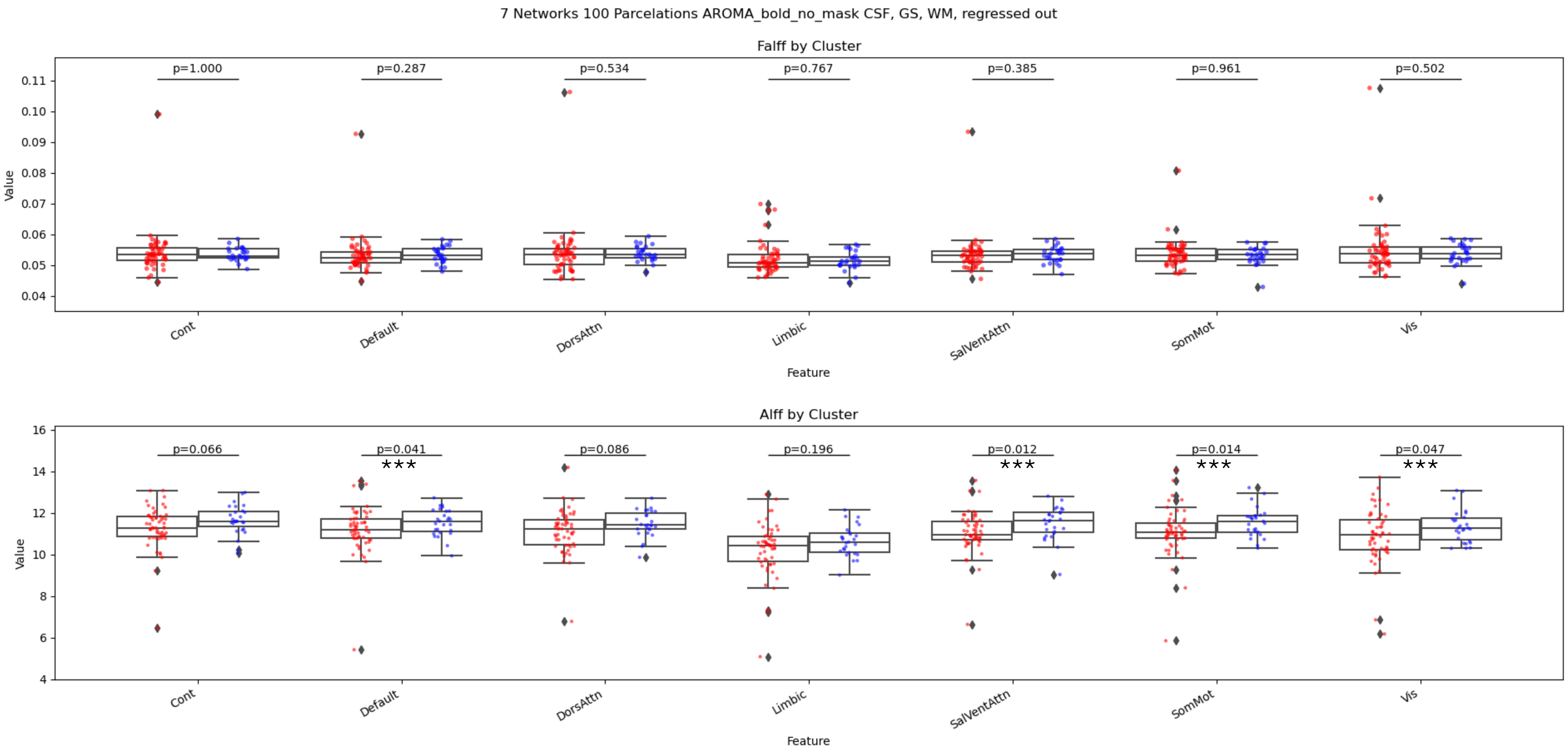


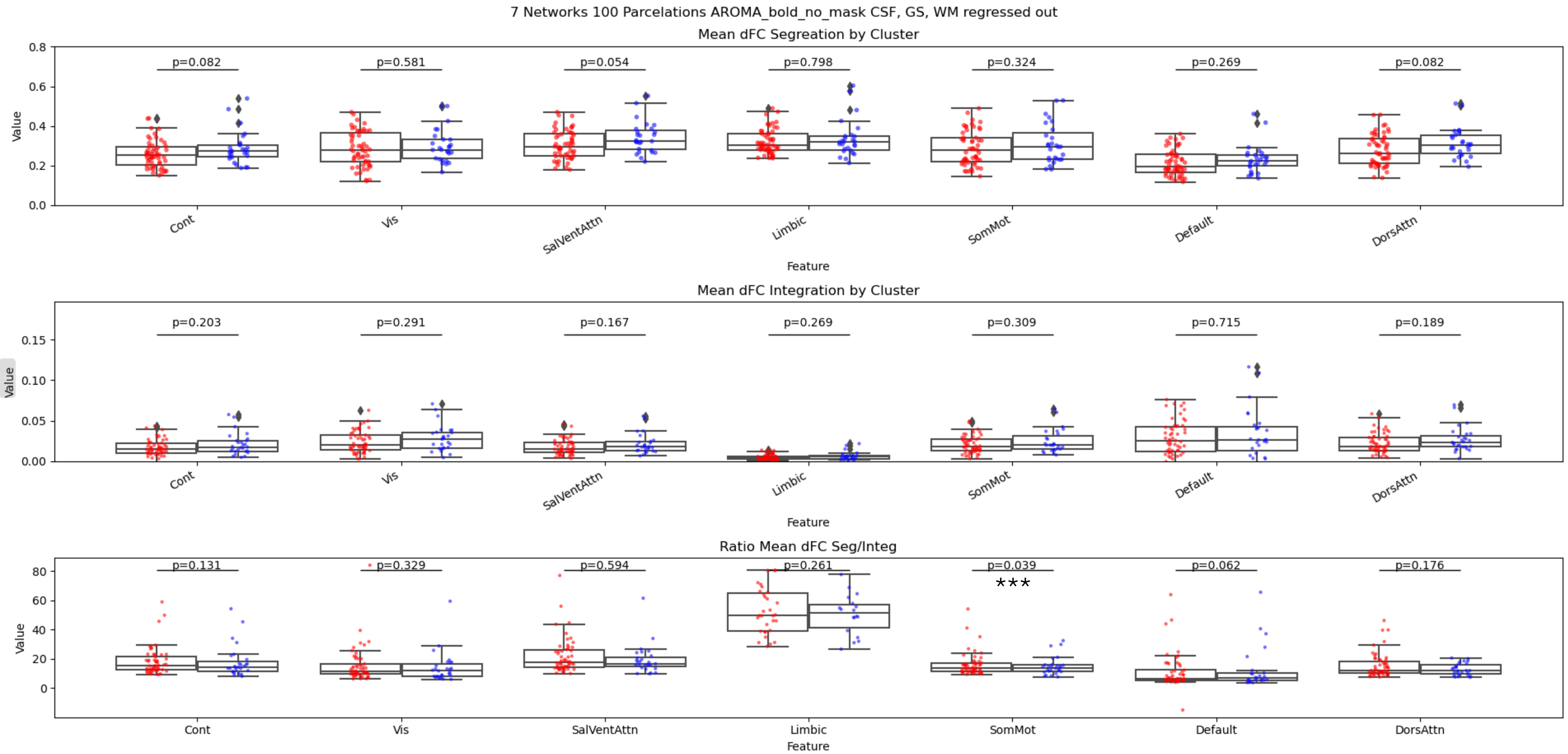
7 Networks 100 Parcellations AROMA no mask CSF, GS, WM regressed out

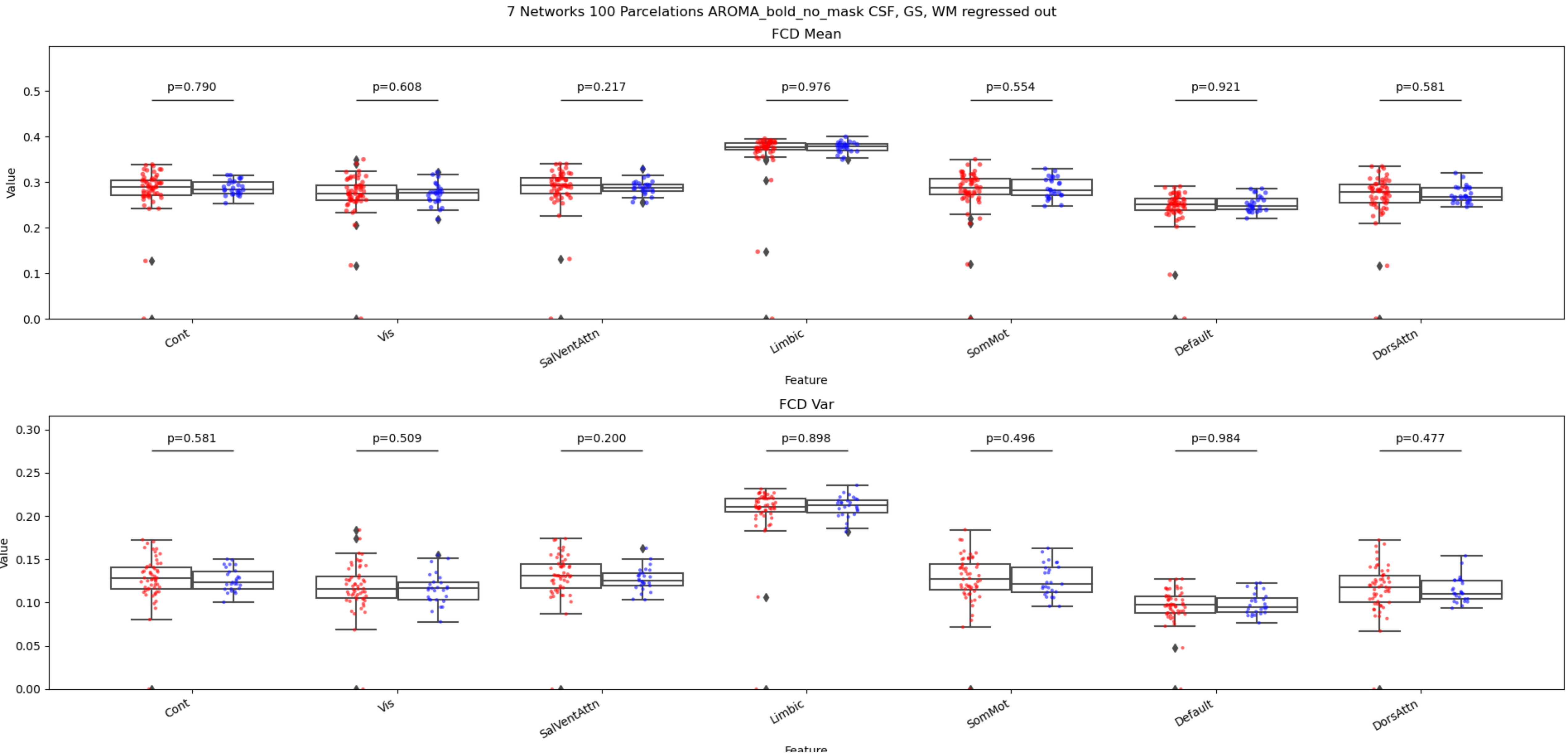


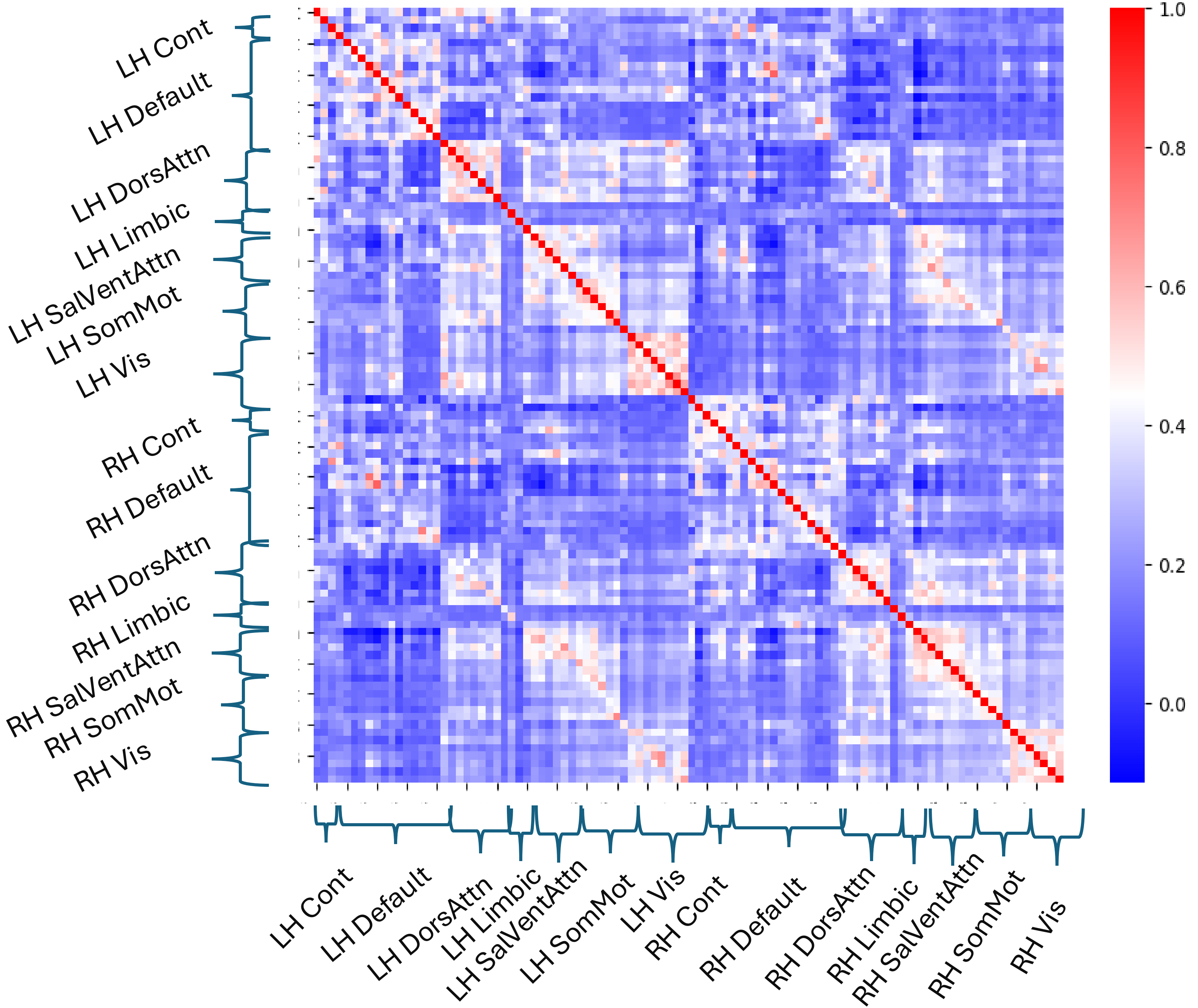


7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM
Segregated FC by Cluster
p=0.369
p=0.820
p=0.047
***
p=0.890
p=0.664
p=0.471
p=0.106
Value
Cont
Vis
SalVentAttn
Limbic
SomMot
Default
DorsAttn
Feature
Integrated FC by Cluster
p=0.396
p=0.547
p=0.248
p=0.601
p=0.686
p=0.945
p=0.447
Value
Feature
Ratio Seg/Integ
p=0.715
p=0.560
p=0.767
p=0.300
p=0.767
p=0.813
p=0.859
Value
Feature

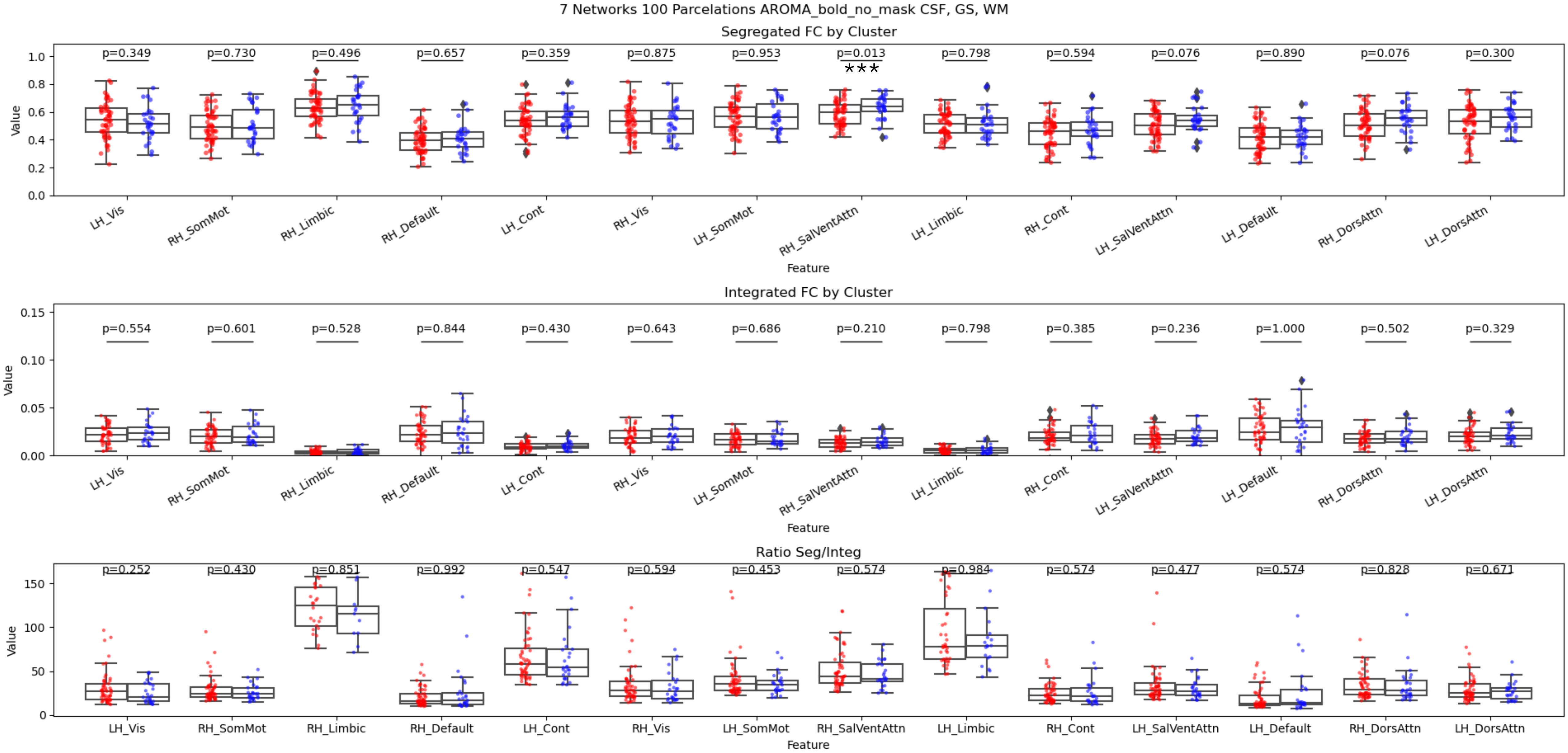

7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, regressed out
Falff by Cluster
p=1.000
p=0.287
p=0.534
p=0.767
p=0.385
p=0.961
p=0.502
Value
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
Cont
Default
DorsAttn
Limbic
SalVentAttn
SomMot
Vis
Feature
Alff by Cluster
p=0.066
p=0.041
***
p=0.086
p=0.196
p=0.012
***
p=0.014
***
p=0.047
***
Value
16
14
12
10
8
6
4
Cont
Default
DorsAttn
Limbic
SalVentAttn
SomMot
Vis
Feature

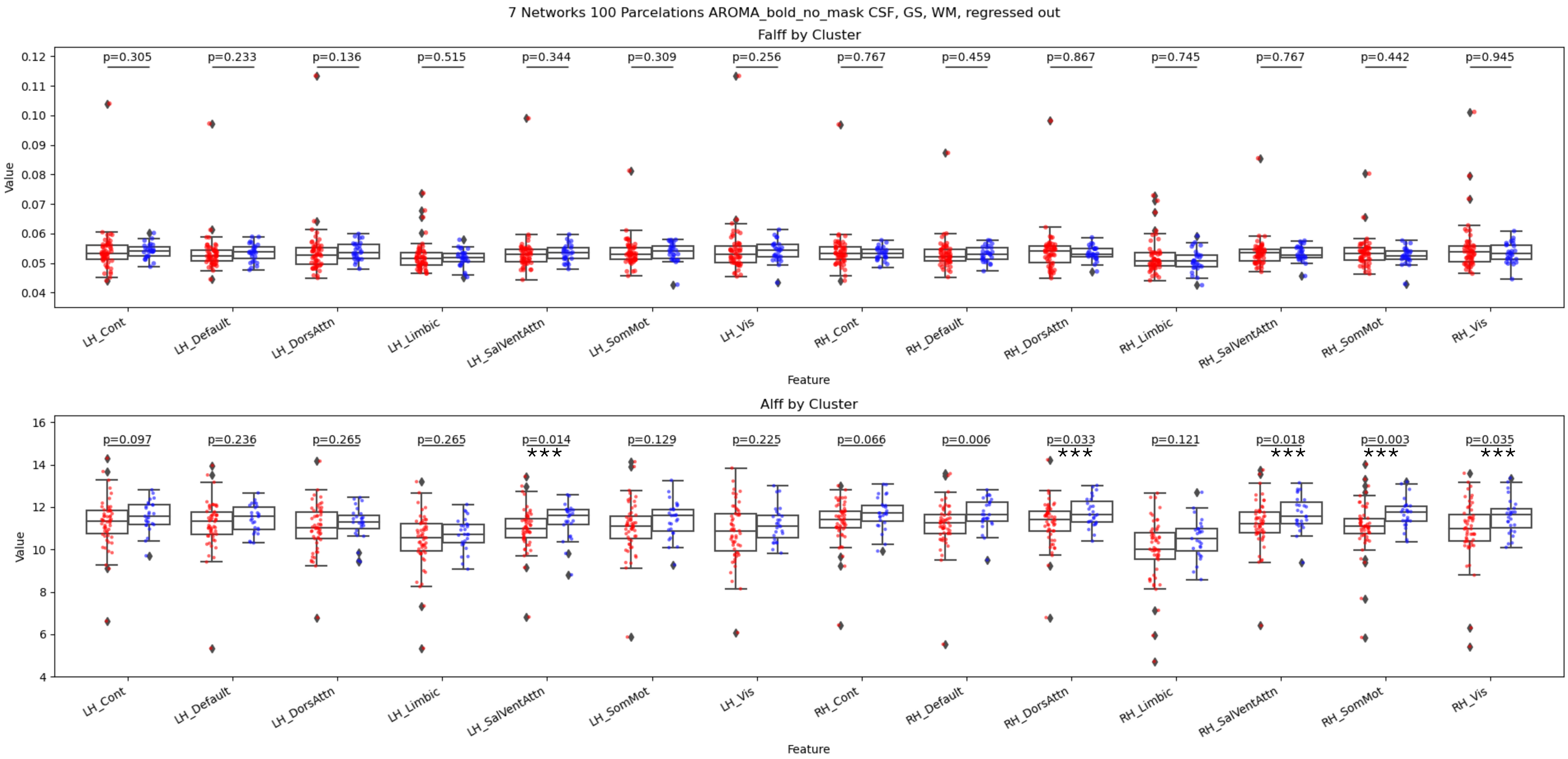

7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out
Mean dFC Segreation by Cluster
p=0.082
p=0.581
p=0.054
p=0.798
p=0.324
p=0.269
p=0.082
Value
Feature
Cont
Vis
SalVentAttn
Limbic
SomMot
Default
DorsAttn
Mean dFC Integration by Cluster
p=0.203
p=0.291
p=0.167
p=0.269
p=0.309
p=0.715
p=0.189
Ratio Mean dFC Seg/Integ
p=0.131
p=0.329
p=0.594
p=0.261
p=0.039
***
p=0.062
p=0.176


7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out
FCD Mean
p=0.790
p=0.608
p=0.217
p=0.976
p=0.554
p=0.921
p=0.581
Value
Cont
Vis
SalVentAttn
Limbic
SomMot
Default
DorsAttn
Feature
FCD Var
p=0.581
p=0.509
p=0.200
p=0.898
p=0.496
p=0.984
p=0.477
Feature

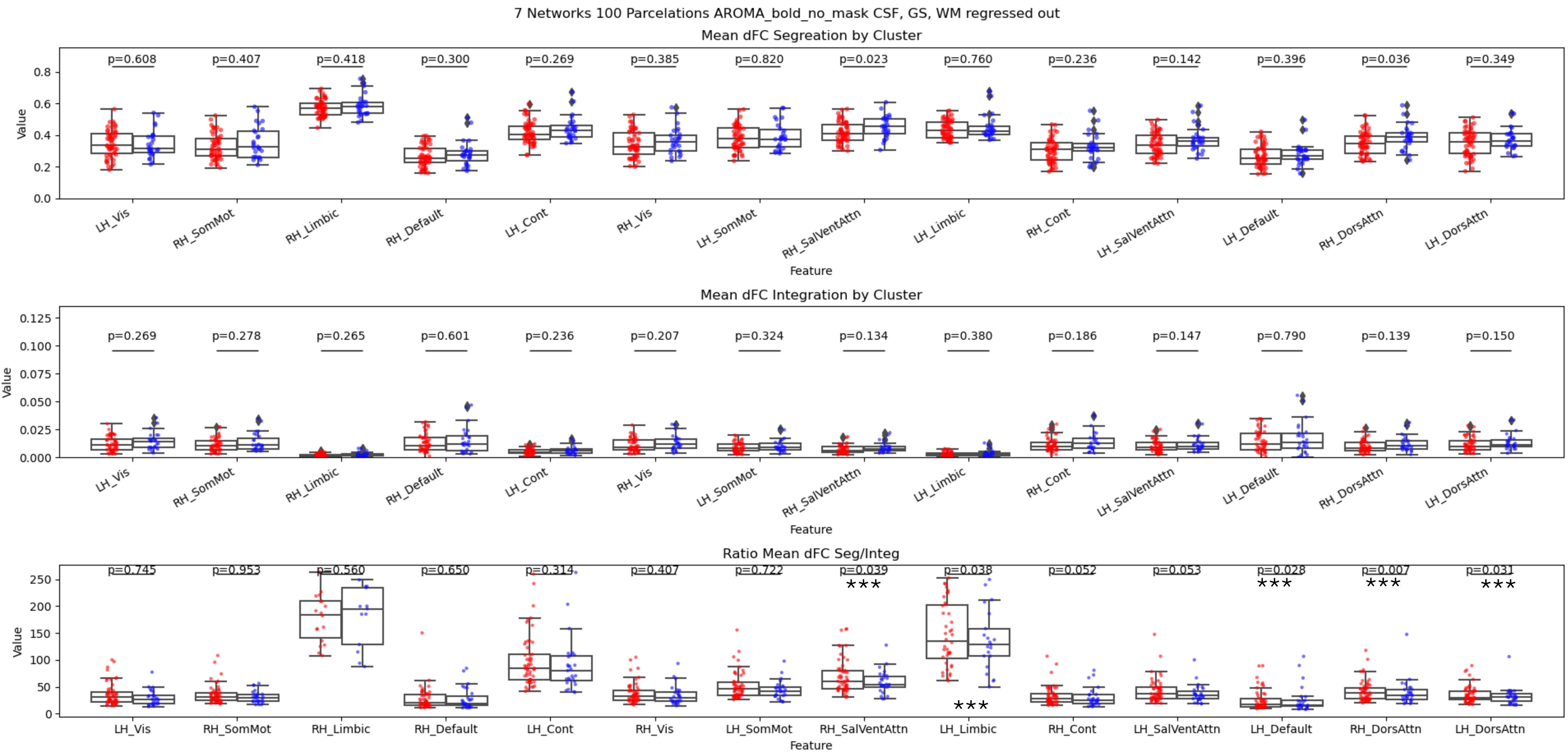


LH Cont
LH Default
LH DorsAttn
LH Limbic
LH SalVentAttn
LH SomMot
LH Vis
RH Cont
RH Default
RH DorsAttn
RH Limbic
RH SalVentAttn
RH SomMot
RH Vis
1.0
0.8
0.6
0.4
0.2
0.0

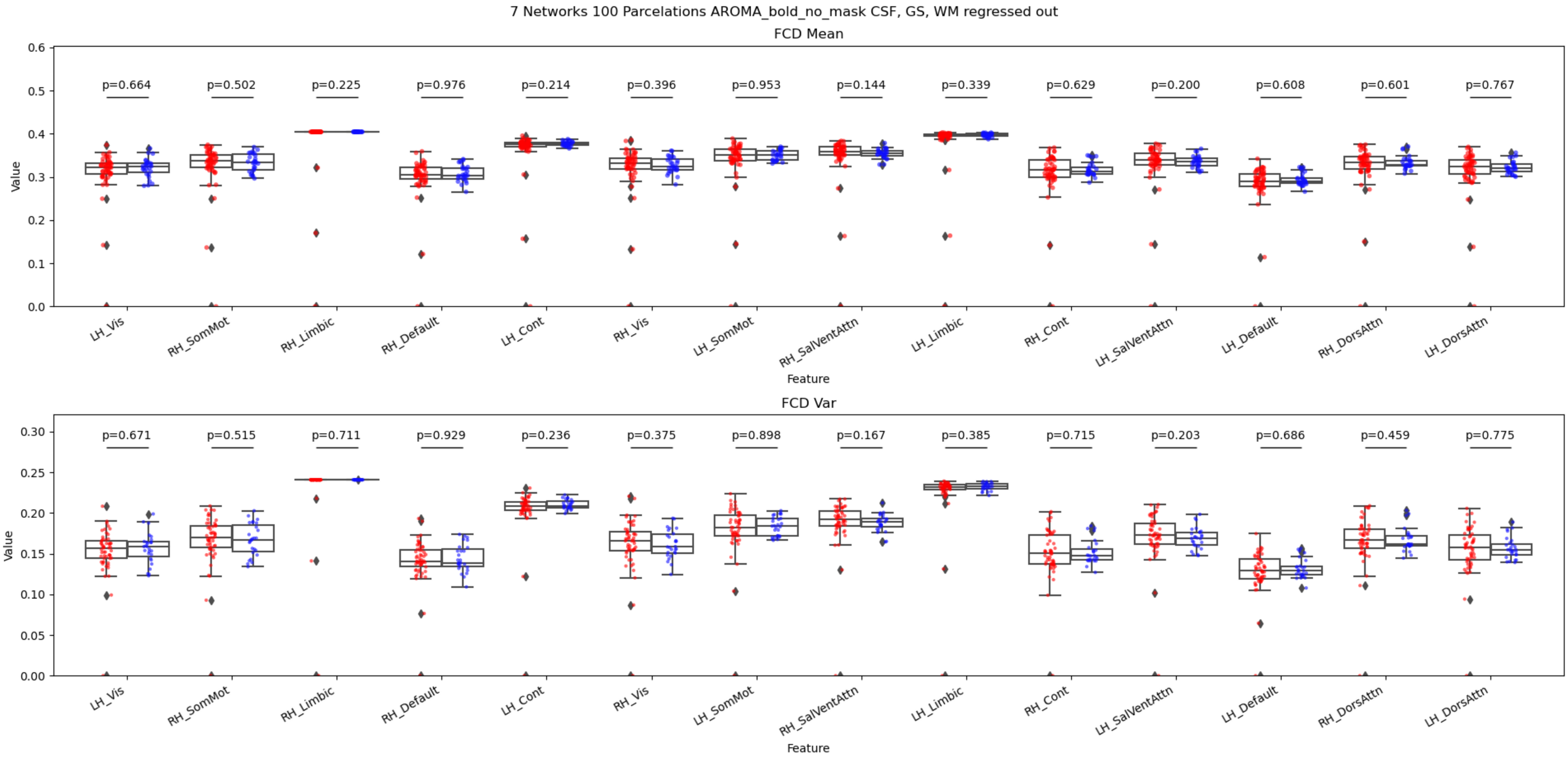

7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM
Segregated FC by Cluster
p=0.349
p=0.730
p=0.496
p=0.657
p=0.359
p=0.875
p=0.953
p=0.013
***
p=0.798
p=0.594
p=0.076
p=0.890
p=0.076
p=0.300
Value
Feature
Integrated FC by Cluster
p=0.554
p=0.601
p=0.528
p=0.844
p=0.430
p=0.643
p=0.686
p=0.210
p=0.798
p=0.385
p=0.236
p=1.000
p=0.502
p=0.329
Value
Feature
Ratio Seg/Integ
p=0.252
p=0.430
p=0.851
p=0.992
p=0.547
p=0.594
p=0.453
p=0.574
p=0.984
p=0.574
p=0.477
p=0.574
p=0.828
p=0.671
Value
LH_Vis
RH_SomMot
RH_Limbic
RH_Default
LH_Cont
RH_Vis
LH_SomMot
RH_SalVentAttn
LH_Limbic
RH_Cont
LH_SalVentAttn
LH_Default
RH_DorsAttn
LH_DorsAttn
Feature



7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, regressed out
Falff by Cluster
Value
Feature
p=0.305
p=0.233
p=0.136
p=0.515
p=0.344
p=0.309
p=0.256
p=0.767
p=0.459
p=0.867
p=0.745
p=0.767
p=0.442
p=0.945
LH_Cont
LH_Default
LH_DorsAttn
LH_Limbic
LH_SalVentAttn
LH_SomMot
LH_Vis
RH_Cont
RH_Default
RH_DorsAttn
RH_Limbic
RH_SalVentAttn
RH_SomMot
RH_Vis
Alff by Cluster
p=0.097
p=0.236
p=0.265
p=0.265
p=0.014
p=0.129
p=0.225
p=0.066
p=0.006
p=0.033
p=0.121
p=0.018
p=0.003
p=0.035



7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out
Mean dFC Segregation by Cluster
p=0.608 p=0.407 p=0.418 p=0.300 p=0.269 p=0.385 p=0.820 p=0.023 p=0.760 p=0.236 p=0.142 p=0.396 p=0.036 p=0.349
Value
LH_Vis RH_SomMot RH_Limbic RH_Default LH_Cont RH_Vis LH_SomMot RH_SalVentAttn LH_Limbic RH_Cont LH_SalVentAttn LH_Default RH_DorsAttn LH_DorsAttn
Feature
Mean dFC Integration by Cluster
p=0.269 p=0.278 p=0.265 p=0.601 p=0.236 p=0.207 p=0.324 p=0.134 p=0.380 p=0.186 p=0.147 p=0.790 p=0.139 p=0.150
Value
LH_Vis RH_SomMot RH_Limbic RH_Default LH_Cont RH_Vis LH_SomMot RH_SalVentAttn LH_Limbic RH_Cont LH_SalVentAttn LH_Default RH_DorsAttn LH_DorsAttn
Feature
Ratio Mean dFC Seg/Integ
p=0.745 p=0.953 p=0.560 p=0.650 p=0.314 p=0.407 p=0.722 p=0.039 p=0.038 p=0.052 p=0.053 p=0.028 p=0.007 p=0.031
***
Value
LH_Vis RH_SomMot RH_Limbic RH_Default LH_Cont RH_Vis LH_SomMot RH_SalVentAttn LH_Limbic RH_Cont LH_SalVentAttn LH_Default RH_DorsAttn LH_DorsAttn
Feature



7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out
FCD Mean
Value
Feature
p=0.664
p=0.502
p=0.225
p=0.976
p=0.214
p=0.396
p=0.953
p=0.144
p=0.339
p=0.629
p=0.200
p=0.608
p=0.601
p=0.767
LH_Vis
RH_SomMot
RH_Limbic
RH_Default
LH_Cont
RH_Vis
LH_SomMot
RH_SalVentAttn
LH_Limbic
RH_Cont
LH_SalVentAttn
LH_Default
RH_DorsAttn
LH_DorsAttn
FCD Var
p=0.671
p=0.515
p=0.711
p=0.929
p=0.236
p=0.375
p=0.898
p=0.167
p=0.385
p=0.715
p=0.203
p=0.686
p=0.459
p=0.775

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An improved framework for confound regression and filtering for control of motion artifact in the preprocessing of resting-state functional connectivity data

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Resting-state

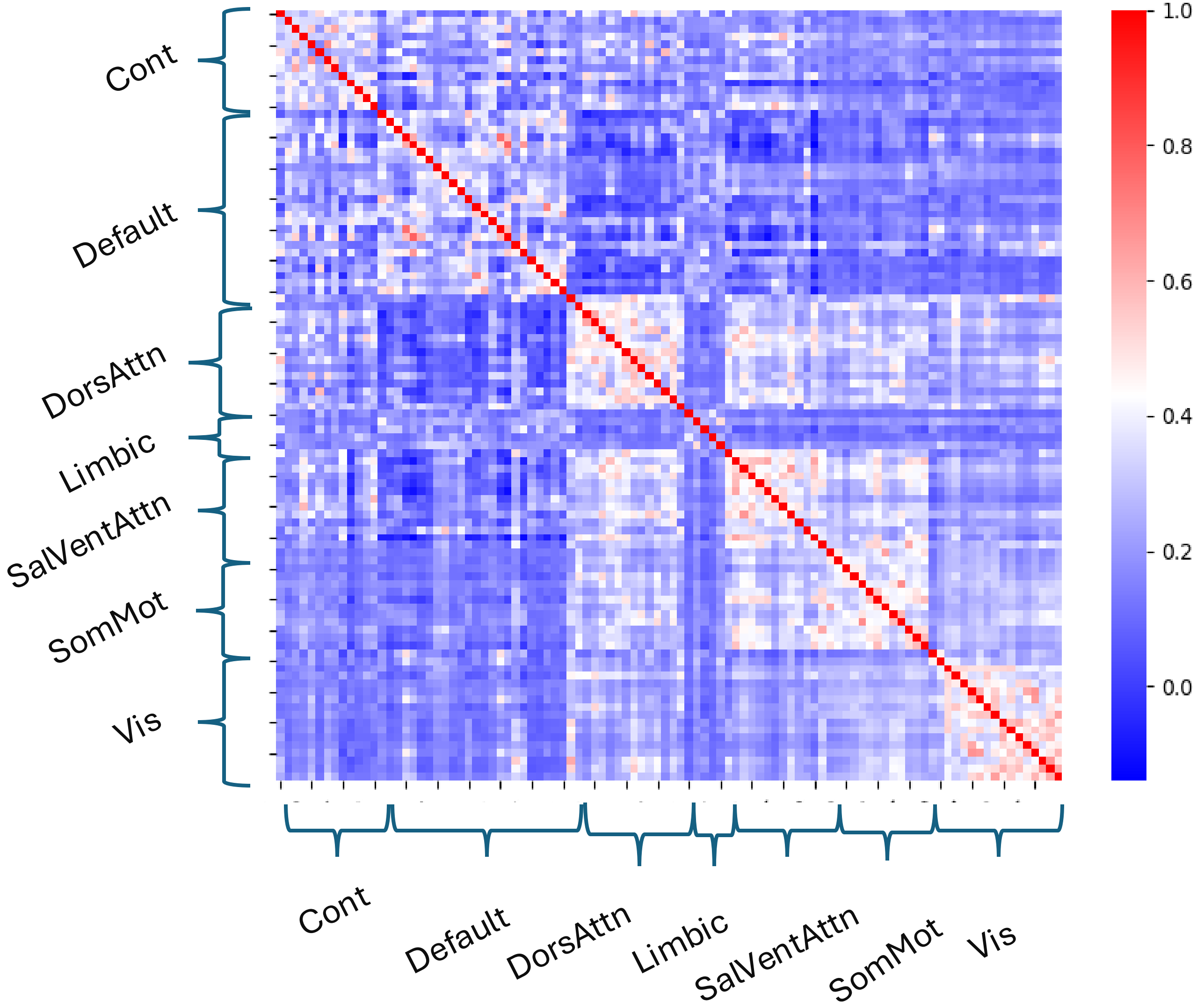
ABSTRACT

Several recent reports in large, independent samples have demonstrated the influence of motion artifact on resting-state functional connectivity MRI (rsfc-MRI). Standard rsfc-MRI preprocessing typically includes regression of confounding signals and band-pass filtering. However, substantial heterogeneity exists in how these techniques are implemented across studies, and no prior study has examined the effect of differing approaches for the control of motion-induced artifacts. To better understand how in-scanner head motion affects rsfc-MRI data, we describe the spatial, temporal, and spectral characteristics of motion artifacts in a sample of 348 adolescents. Analyses utilize a novel approach for describing head motion on a voxelwise basis. Next, we systematically evaluate the efficacy of a range of confound regression and filtering techniques for the control of motion-induced artifacts. Results reveal that the effectiveness of preprocessing procedures on the control of motion is heterogeneous, and that improved preprocessing provides a substantial benefit beyond typical procedures. These results demonstrate that the effect of motion on rsfc-MRI can be substantially attenuated through improved preprocessing procedures, but not completely removed.

Motion parameters + temporal derivatives and quadratic terms for the three global signals (csf, white_matter and global_signal)
36-parameter denoising strategy proposed

So we do AROMA denoising + 12 more parameters

The article argues that motion artifacts correlate with spectrum of low frequency oscillation, so the significance might be due to motion artifacts



Cont
Default
DorsAttn
Limbic
SalVentAttn
SomMot
Vis
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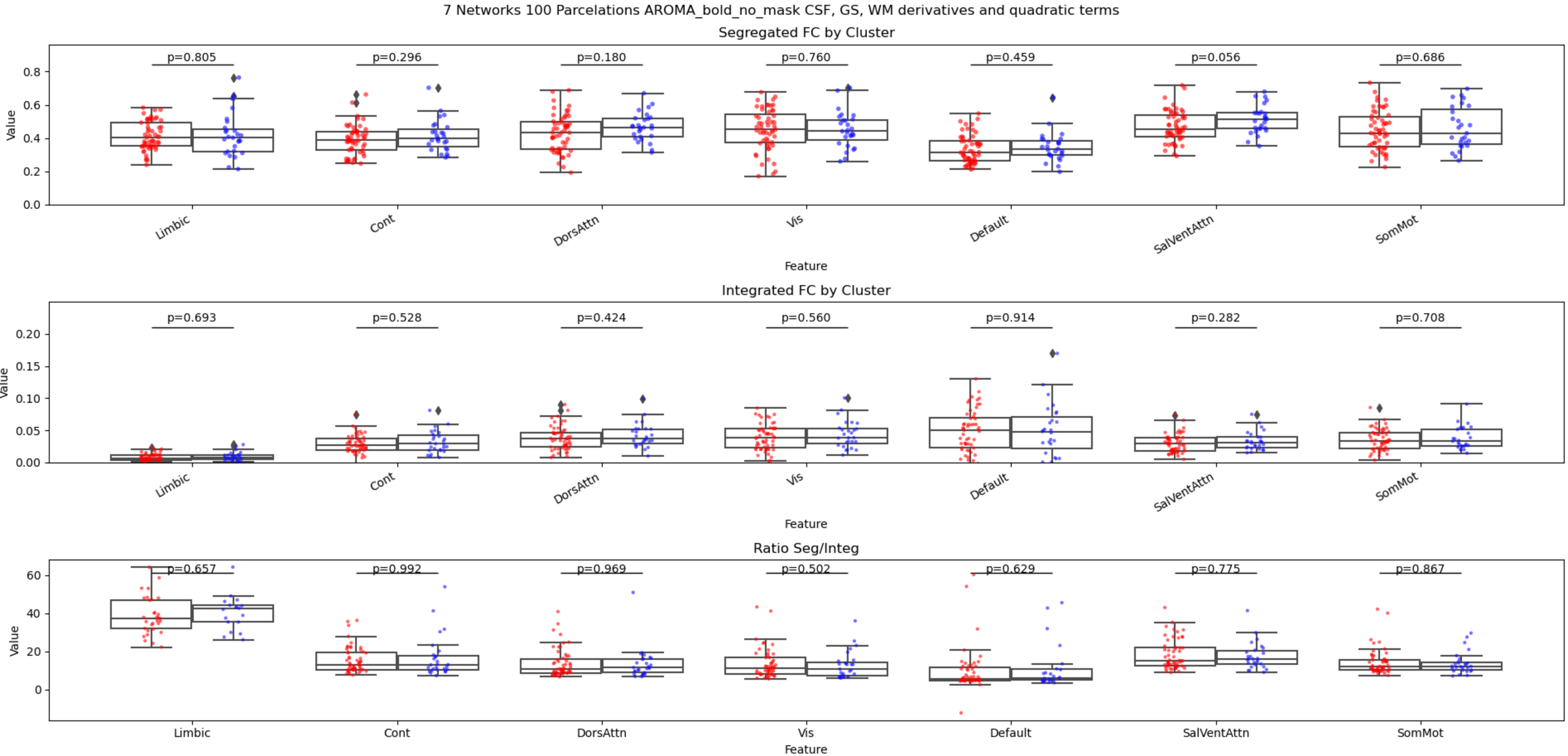
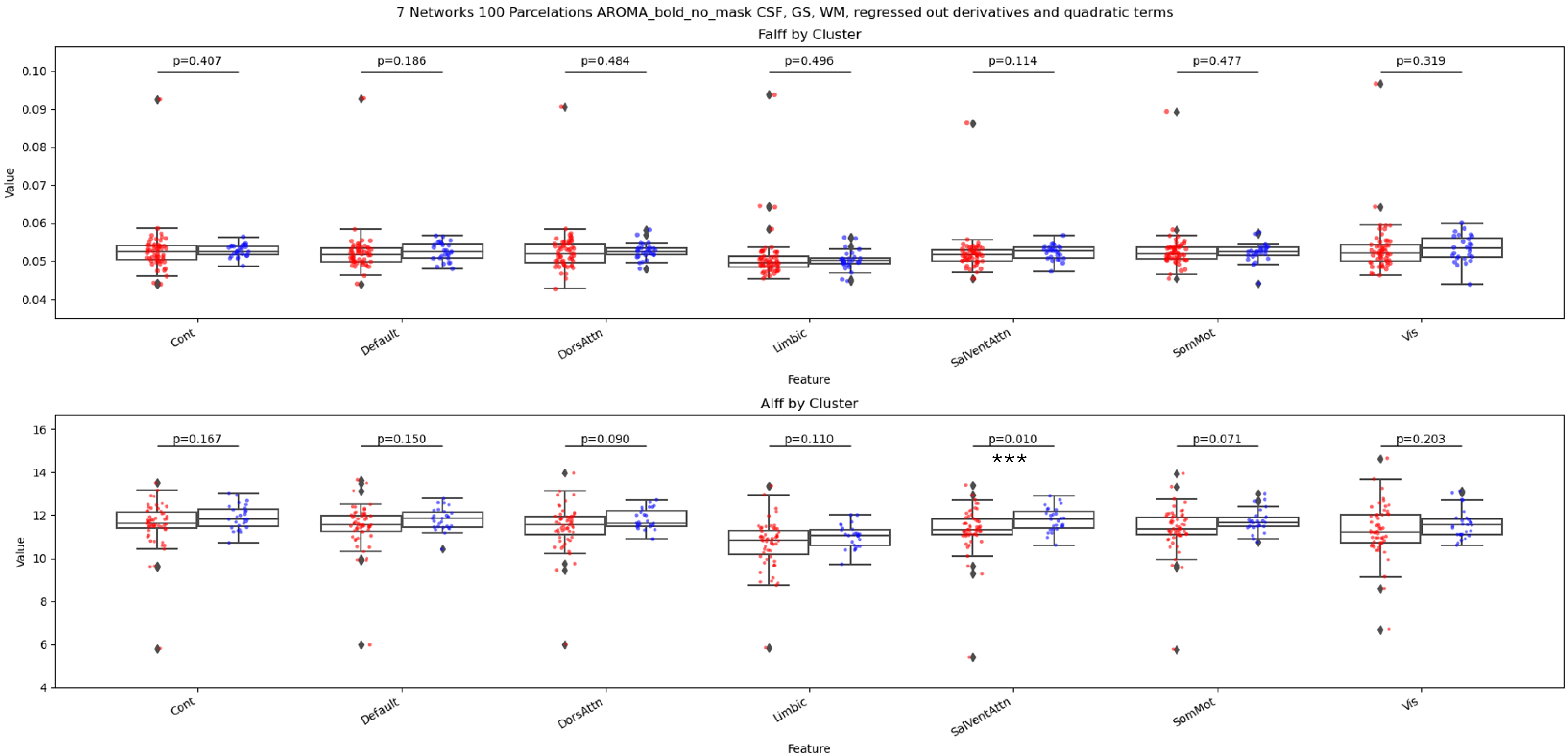
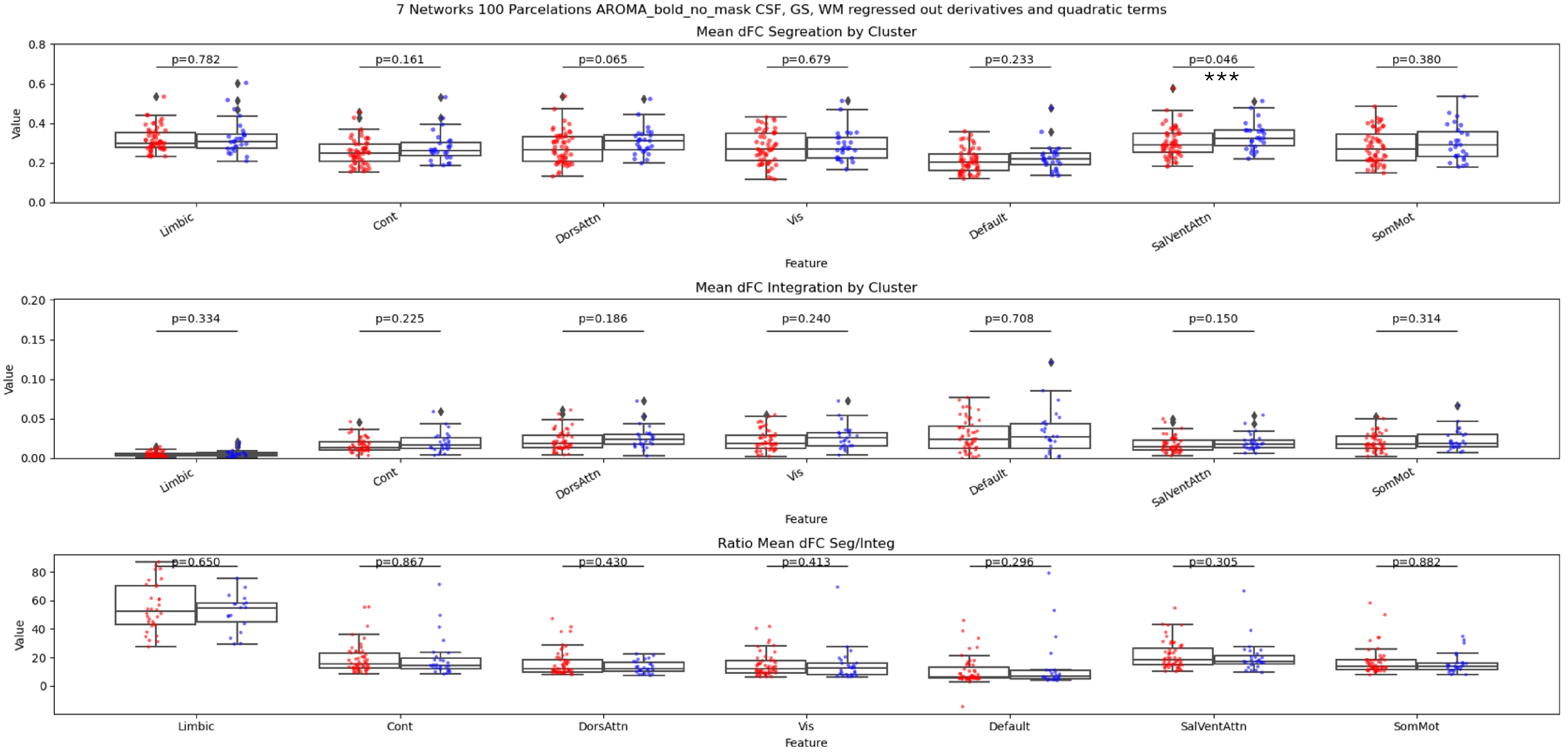
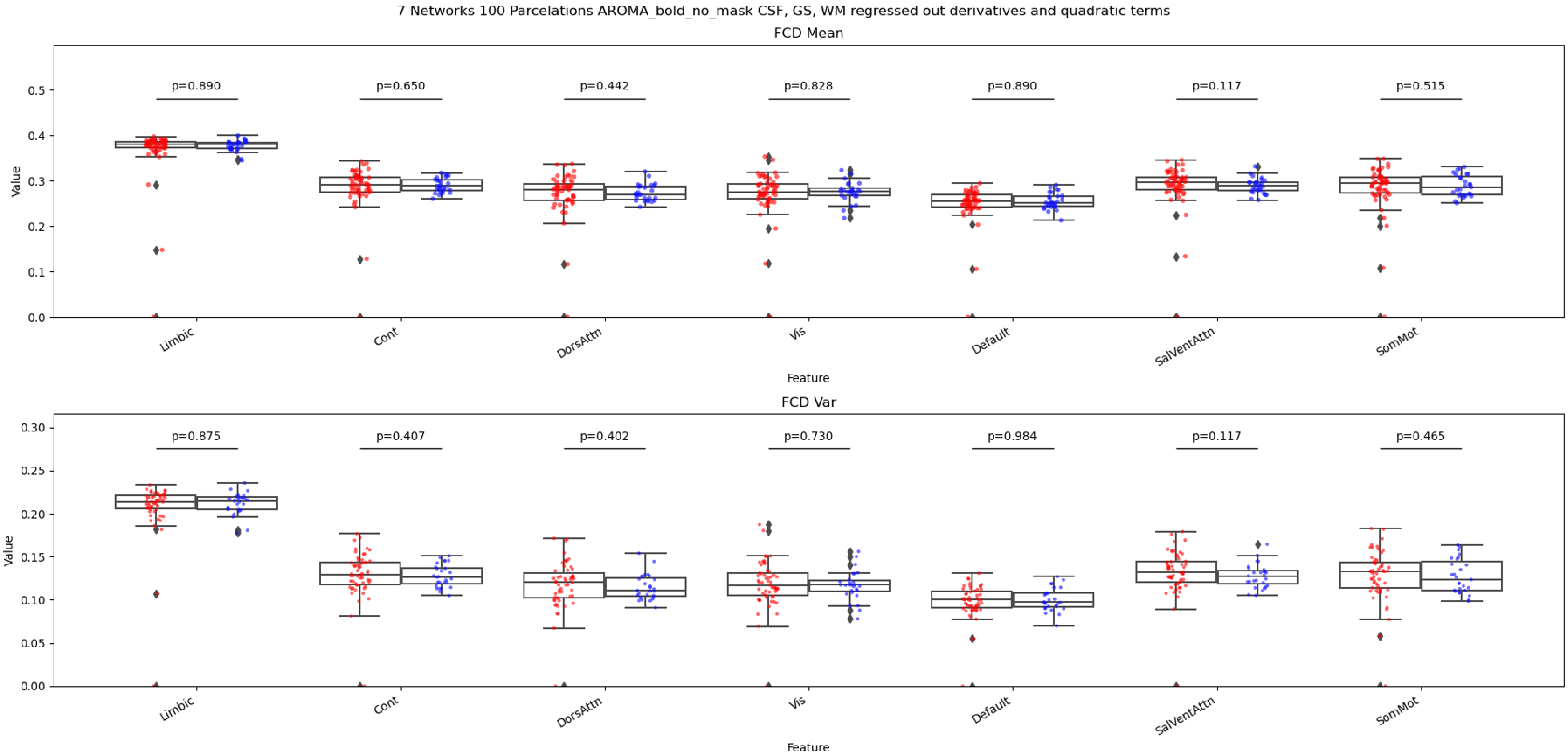
0.4
0.2
0.0

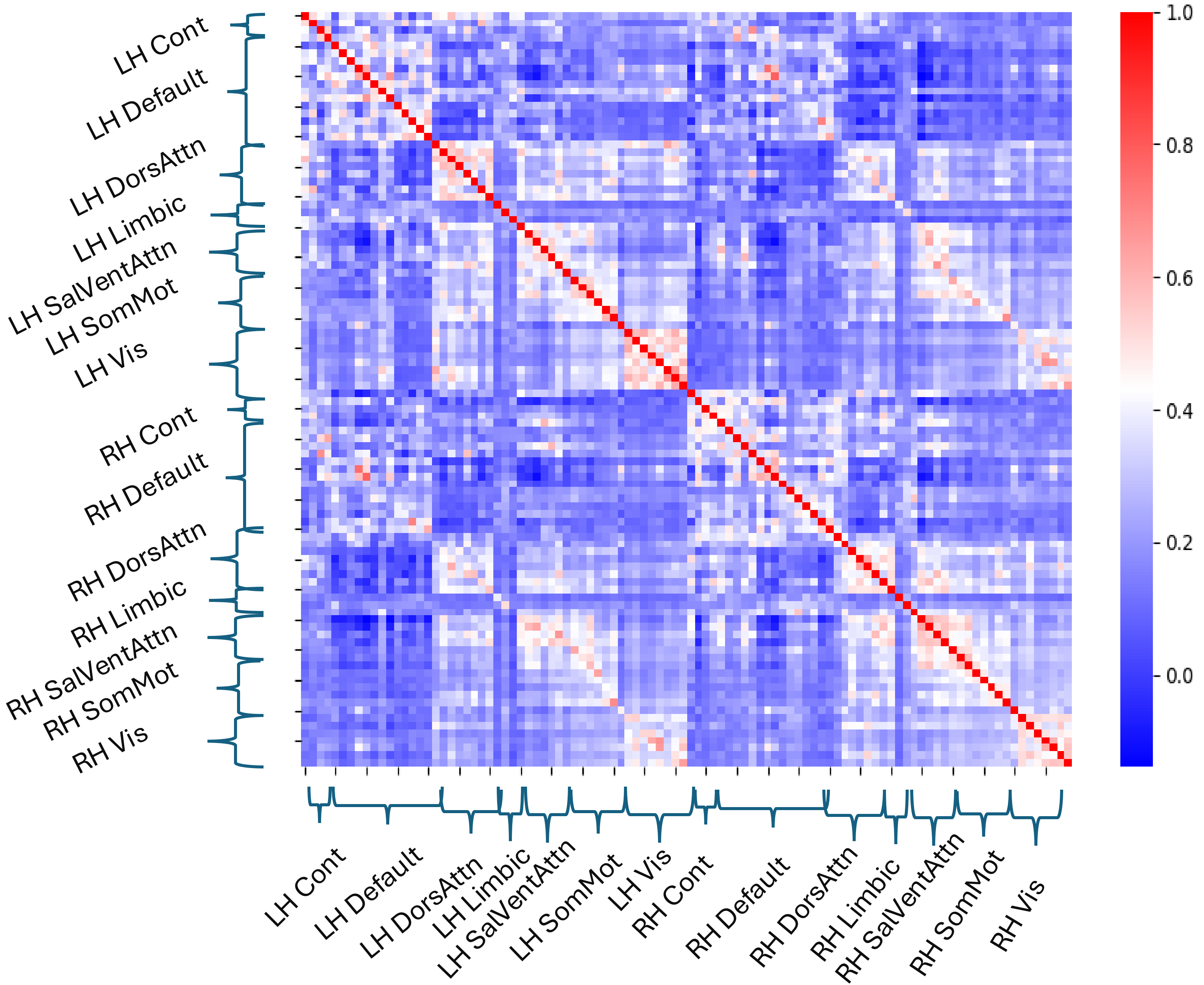

7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM derivatives and quadratic terms
Segregated FC by Cluster
p=0.805
p=0.296
p=0.180
p=0.760
p=0.459
p=0.056
p=0.686
Value
Limbic
Cont
DorsAttn
Vis
Default
SalVentAttn
SomMot
Feature
Integrated FC by Cluster
p=0.693
p=0.528
p=0.424
p=0.560
p=0.914
p=0.282
p=0.708
Ratio Seg/Integ
p=0.657
p=0.992
p=0.969
p=0.502
p=0.629
p=0.775
p=0.867

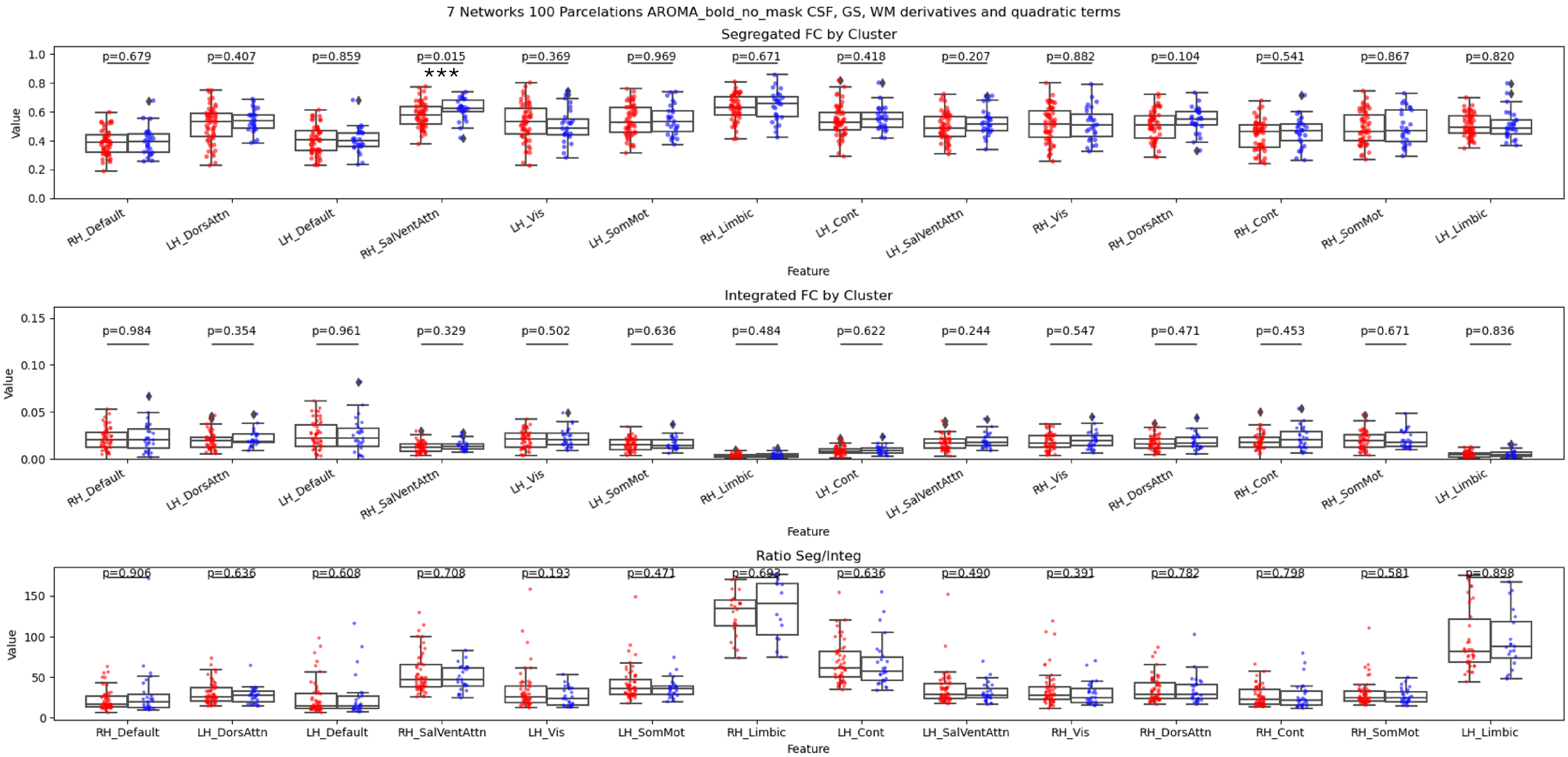

7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, regressed out derivatives and quadratic terms
Falff by Cluster
p=0.407
p=0.186
p=0.484
p=0.496
p=0.114
p=0.477
p=0.319
Value
Cont
Default
DorsAttn
Limbic
SalVentAttn
SomMot
Vis
Feature
Alff by Cluster
p=0.167
p=0.150
p=0.090
p=0.110
p=0.010
***
p=0.071
p=0.203
Value
Cont
Default
DorsAttn
Limbic
SalVentAttn
SomMot
Vis
Feature


7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out derivatives and quadratic terms
Mean dFC Segregation by Cluster
p=0.782
p=0.161
p=0.065
p=0.679
p=0.233
p=0.046
***
p=0.380
Value
Limbic
Cont
DorsAttn
Vis
Default
SalVentAttn
SomMot
Feature
Mean dFC Integration by Cluster
p=0.334
p=0.225
p=0.186
p=0.240
p=0.708
p=0.150
p=0.314
Value
Limbic
Cont
DorsAttn
Vis
Default
SalVentAttn
SomMot
Feature
Ratio Mean dFC Seg/Integ
p=0.650
p=0.867
p=0.430
p=0.413
p=0.296
p=0.305
p=0.882
Value
Limbic
Cont
DorsAttn
Vis
Default
SalVentAttn
SomMot
Feature

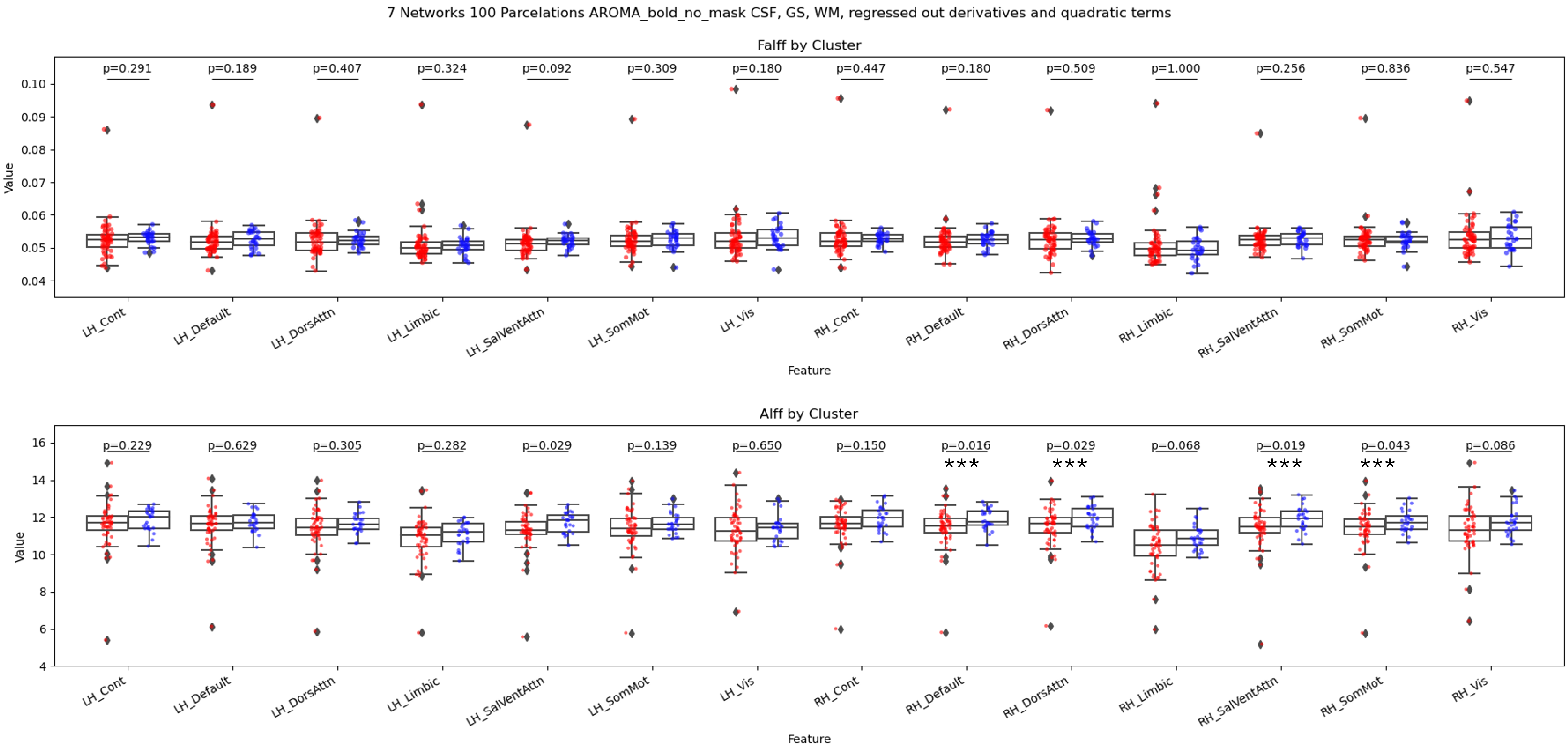


7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out derivatives and quadratic terms
FCD Mean
p=0.890
p=0.650
p=0.442
p=0.828
p=0.890
p=0.117
p=0.515
Value
0.0
0.1
0.2
0.3
0.4
0.5
Limbic
Cont
DorsAttn
Vis
Default
SalVentAttn
SomMot
Feature
FCD Var
p=0.875
p=0.407
p=0.402
p=0.730
p=0.984
p=0.117
p=0.465
Value
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Limbic
Cont
DorsAttn
Vis
Default
SalVentAttn
SomMot
Feature

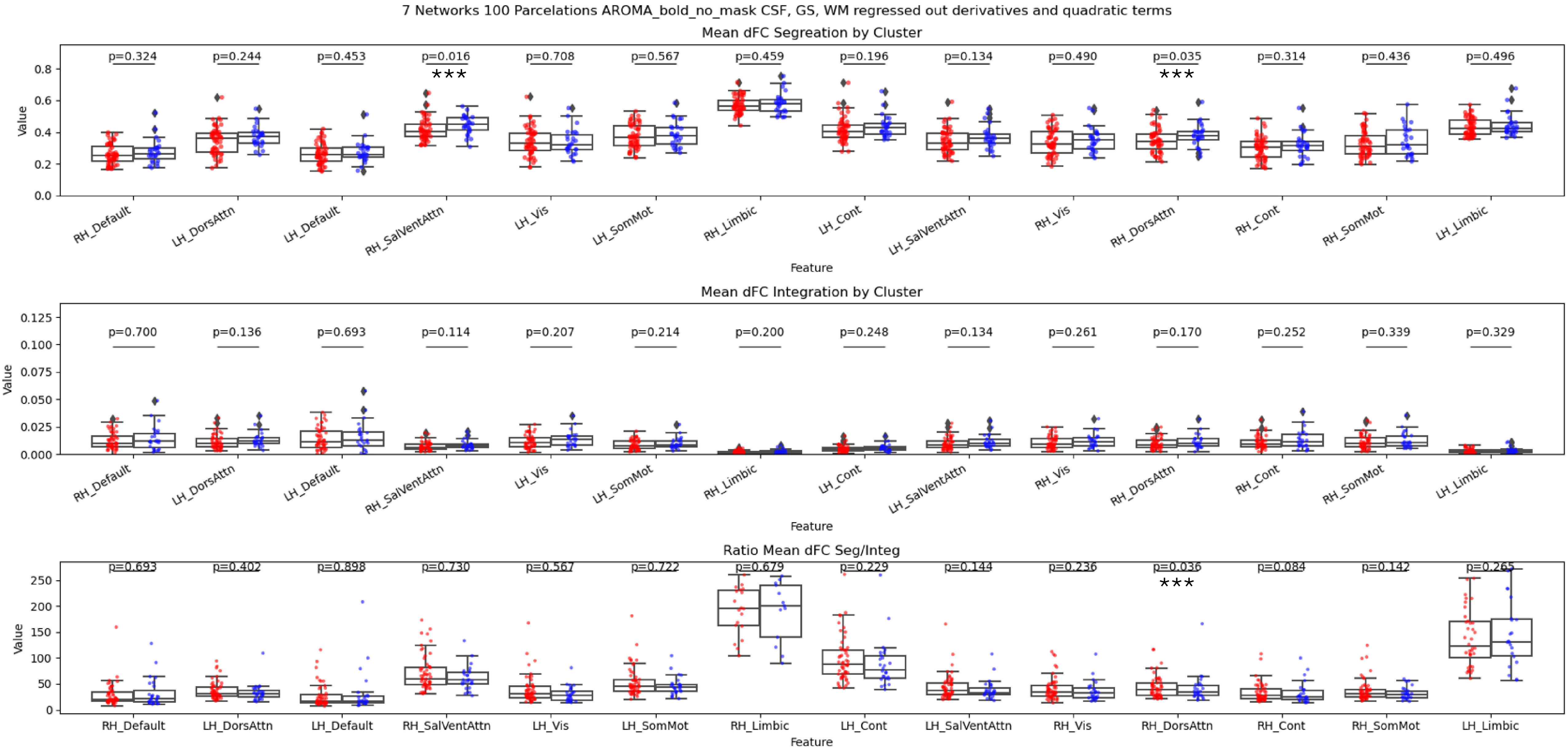

LH Cont
LH Default
LH DorsAttn
LH Limbic
LH SalVentAttn
LH SomMot
LH Vis
RH Cont
RH Default
RH DorsAttn
RH Limbic
RH SalVentAttn
RH SomMot
RH Vis
1.0
0.8
0.6
0.4
0.2
0.0



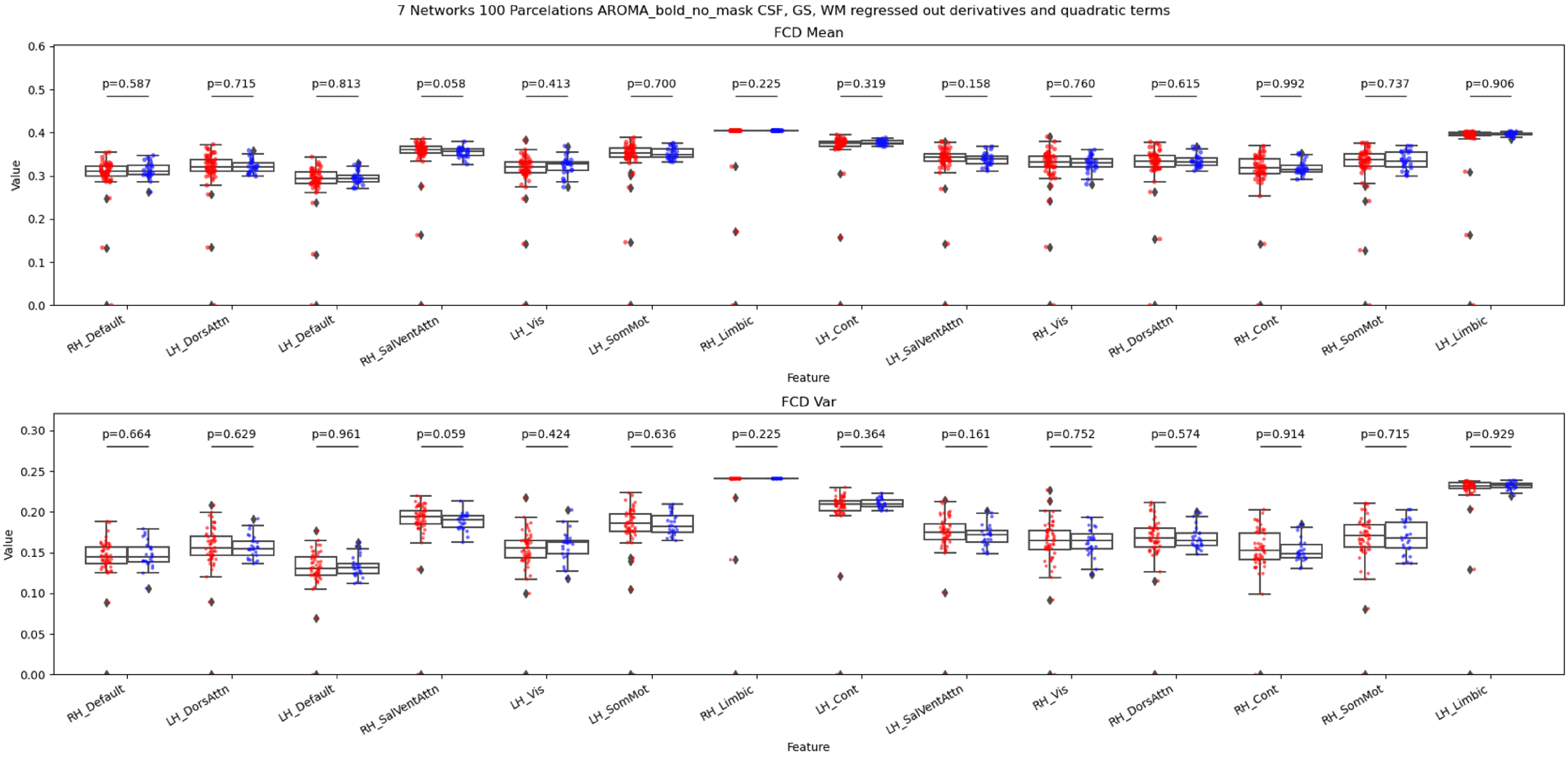
7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM derivatives and quadratic terms
Segregated FC by Cluster
p=0.679
p=0.407
p=0.859
p=0.015
***
p=0.369
p=0.969
p=0.671
p=0.418
p=0.207
p=0.882
p=0.104
p=0.541
p=0.867
p=0.820
Value
Feature
Integrated FC by Cluster
p=0.984
p=0.354
p=0.961
p=0.329
p=0.502
p=0.636
p=0.484
p=0.622
p=0.244
p=0.547
p=0.471
p=0.453
p=0.671
p=0.836
Ratio Seg/Integ
p=0.906
p=0.636
p=0.608
p=0.708
p=0.193
p=0.471
p=0.693
p=0.636
p=0.490
p=0.391
p=0.782
p=0.798
p=0.581
p=0.898
RH_Default
LH_DorsAttn
LH_Default
RH_SalVentAttn
LH_Vis
LH_SomMot
RH_Limbic
LH_Cont
LH_SalVentAttn
RH_Vis
RH_DorsAttn
RH_Cont
RH_SomMot
LH_Limbic



7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM, regressed out derivatives and quadratic terms
Falff by Cluster
p=0.291
p=0.189
p=0.407
p=0.324
p=0.092
p=0.309
p=0.180
p=0.447
p=0.180
p=0.509
p=1.000
p=0.256
p=0.836
p=0.547
Value
Feature
LH_Cont
LH_Default
LH_DorsAttn
LH_Limbic
LH_SalVentAttn
LH_SomMot
LH_Vis
RH_Cont
RH_Default
RH_DorsAttn
RH_Limbic
RH_SalVentAttn
RH_SomMot
RH_Vis
Alff by Cluster
p=0.229
p=0.629
p=0.305
p=0.282
p=0.029
p=0.139
p=0.650
p=0.150
p=0.016
p=0.029
p=0.068
p=0.019
p=0.043
p=0.086
***
Value
Feature



7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out derivatives and quadratic terms
Mean dFC Segregation by Cluster
p=0.324
p=0.244
p=0.453
p=0.016
***
p=0.708
p=0.567
p=0.459
p=0.196
p=0.134
p=0.490
p=0.035
***
p=0.314
p=0.436
p=0.496
Value
RH_Default
LH_DorsAttn
LH_Default
RH_SalVentAttn
LH_Vis
LH_SomMot
RH_Limbic
LH_Cont
LH_SalVentAttn
RH_Vis
RH_DorsAttn
RH_Cont
RH_SomMot
LH_Limbic
Feature
Mean dFC Integration by Cluster
p=0.700
p=0.136
p=0.693
p=0.114
p=0.207
p=0.214
p=0.200
p=0.248
p=0.134
p=0.261
p=0.170
p=0.252
p=0.339
p=0.329
Value
Feature
Ratio Mean dFC Seg/Integ
p=0.693
p=0.402
p=0.898
p=0.730
p=0.567
p=0.722
p=0.679
p=0.229
p=0.144
p=0.236
p=0.036
***
p=0.084
p=0.142
p=0.265
Value
Feature



7 Networks 100 Parcelations AROMA_bold_no_mask CSF, GS, WM regressed out derivatives and quadratic terms
FCD Mean
p=0.587
p=0.715
p=0.813
p=0.058
p=0.413
p=0.700
p=0.225
p=0.319
p=0.158
p=0.760
p=0.615
p=0.992
p=0.737
p=0.906
Value
RH_Default
LH_DorsAttn
LH_Default
RH_SalVentAttn
LH_Vis
LH_SomMot
RH_Limbic
LH_Cont
LH_SalVentAttn
RH_Vis
RH_DorsAttn
RH_Cont
RH_SomMot
LH_Limbic
Feature
FCD Var
p=0.664
p=0.629
p=0.961
p=0.059
p=0.424
p=0.636
p=0.225
p=0.364
p=0.161
p=0.752
p=0.574
p=0.914
p=0.715
p=0.929
Value
RH_Default
LH_DorsAttn
LH_Default
RH_SalVentAttn
LH_Vis
LH_SomMot
RH_Limbic
LH_Cont
LH_SalVentAttn
RH_Vis
RH_DorsAttn
RH_Cont
RH_SomMot
LH_Limbic
Feature